

The  
appli  
catio  
n of  
conti  
nuo  
us  
impr  
ove  
men  
t  
and  
lean  
six  
sigm  
a-  
DMA  
IC  
met  
hod  
ologi  
es in  
phar  
mac  
eutic  
al  
indu  
stry

M  
o  
h  
a  
m  
m  
a  
d  
  
D  
  
A  
L  
-  
T  
a  
h  
a  
t  
1  
  
a  
n  
d  
  
M  
a  
i  
s  
a  
  
A  
  
N

|                                                                                   |      |
|-----------------------------------------------------------------------------------|------|
|                                                                                   | 2025 |
| sour <sup>2</sup>                                                                 | A    |
|  | r    |
|                                                                                   | ti   |
|                                                                                   | c    |
|                                                                                   | l    |
|                                                                                   | e    |
| Abstract                                                                          | r    |
|                                                                                   | e    |
| Human                                                                             | u    |
| Systems                                                                           | s    |
| Managemen                                                                         | e    |
| t                                                                                 | g    |
| 2                                                                                 | u    |
| 0                                                                                 | i    |
| 2                                                                                 | d    |
| 5                                                                                 | e    |
| ,                                                                                 | li   |
|                                                                                   | n    |
| V                                                                                 | e    |
| o                                                                                 | s    |
| l                                                                                 | :    |
| .                                                                                 | s    |
|                                                                                   | a    |
| 0                                                                                 | g    |
| (                                                                                 | e    |
| 0                                                                                 | p    |
| )                                                                                 | u    |
|                                                                                   | b    |
| 1                                                                                 | .    |
| -                                                                                 | c    |
| 1                                                                                 | o    |
| 6                                                                                 | m    |
|                                                                                   | /j   |
| ©                                                                                 | o    |
|                                                                                   | u    |
| T                                                                                 | r    |
| h                                                                                 | n    |
| e                                                                                 | a    |
|                                                                                   | ls   |
| A                                                                                 | -    |
| u                                                                                 | p    |
| t                                                                                 | e    |
| h                                                                                 | r    |
| o                                                                                 | m    |
| r                                                                                 | is   |
| (                                                                                 | si   |
| s                                                                                 | o    |
| )                                                                                 | n    |
|                                                                                   | s    |
|                                                                                   | D    |
|                                                                                   | O    |

Background: The pharmaceutical industry is continuously striving to enhance operational efficiency. Variability in production, particularly during the tablet compression phase, presents significant challenges to maintaining consistent product quality. The Six Sigma DMAIC methodology has emerged as an effective approach for addressing these process inefficiencies.

Objective: This paper aims to apply the Six Sigma DMAIC methodology to improve the tablet compression process for AMOF625 tablets, with the goal of reducing variability and enhancing overall product quality.

Methods: The DMAIC framework was employed in the following phases: Define: Critical areas contributing to nonconformance were identified. Measure: Eight months of historical data were analyzed to identify patterns of variability in the process. Analyze: Rootcause analysis was conducted using statistical tools, including control charts and fishbone diagrams, to pinpoint the sources of variability. Improve: Solutions such as automating procedures and enhancing worker training were implemented to address the identified issues. And Control: To ensure sustained improvements, controls including continuous monitoring systems were established.

Results: The implementation of the Six Sigma DMAIC framework led to substantial improvements in process capability. Notably, sigma levels increased from 3.05 to 5.4 in the powder stage, reflecting a significant reduction in variability. Additionally, product quality metrics showed considerable improvement.

Conclusions: The application of Six Sigma effectively reduced process variability and enhanced product quality in the tablet production process. This study highlights the importance of Six Sigma principles in driving continuous improvement within the pharmaceutical manufacturing sector, demonstrating its potential for achieving sustained quality enhancements.

10.1177/016  
7253325136  
5021  
[journals.sag  
epub.com/h  
ome/hsy](https://journals.sagepub.com/home/hsy)



im |  
pro n  
ve t  
me t  
nt, r  
pro o  
ces d  
s u  
op c  
tim  
iza t  
tio i  
n o  
Rec n  
eive T  
d: h  
20 e  
Nov p  
emb h  
er a  
202 r  
4; m  
revis  
ed: a  
30 c  
June e  
202 u  
5; ti  
acce c  
pted a  
: 1 l  
July i  
202 n  
5 d  
u  
s  
tr  
y  
,  
li  
k  
e  
m  
a  
n  
y  
o  
t  
h

|    |    |
|----|----|
| e  | c  |
| r  | r  |
| s, | e  |
| i  | a  |
| s  | s  |
| b  | i  |
| e  | n  |
| i  | g  |
| n  | d  |
| g  | e  |
| r  | m  |
| e  | a  |
| s  | n  |
| h  | d  |
| a  | s  |
| p  | f  |
| e  | o  |
| d  | r  |
| b  | e  |
| y  | f  |
| r  | fi |
| a  | c  |
| p  | i  |
| i  | e  |
| d  | n  |
| l  | c  |
| y  | y  |
| c  | ,  |
| h  | a  |
| a  | d  |
| n  | a  |
| g  | p  |
| i  | t  |
| n  | a  |
| g  | b  |
| m  | il |
| a  | it |
| r  | y  |
| k  | ,  |
| e  | a  |
| t  | n  |
| c  | d  |
| o  | p  |
| n  | r  |
| d  | o  |
| it | d  |
| i  | u  |
| o  | c  |
| n  | t  |
| s  | c  |
| a  | o  |
| n  | n  |
| d  | s  |
| i  | i  |
| n  | s  |

---

t p  
e r  
n o  
c v  
y e  
. m  
T e  
o n  
m t  
e st  
e r  
t a  
t t  
h e  
e g  
s i  
e e  
c s  
h t  
a h  
ll a  
e t  
n s  
g u  
e p  
s, p  
c o  
o rt  
m b  
p o  
a t  
n h  
i o  
e p  
s e  
a r  
r a  
e ti  
t o  
u n  
r a  
n l  
i p  
n e  
g rf  
t o  
o r  
p m  
r a  
o n  
v c  
e e  
n a  
i n  
m d

---

l y  
o i  
n n  
g it  
- s  
t D  
e M  
r A  
m I  
C  
q (  
u D  
a e  
li fi  
t n  
y e  
a ,  
s M  
s e  
u a  
r s  
a u  
n r  
c e  
e ,  
. A  
A n  
m a  
o l  
n y  
g z  
t e  
h ,  
e I  
s m  
e p  
, r  
S o  
i v  
x e  
S ,  
i a  
g n  
m d  
a C  
— o  
p n  
a tr  
rt o  
i l)  
c f  
u o  
l r  
a m  
rl —

---

h r  
a r  
s e  
e d  
s u  
t c  
a i  
b n  
li g  
s v  
h a  
e ri  
d a  
it ti  
s o  
e n  
lf a  
a n  
s d  
a i  
p m  
r p  
a r  
c o  
ti v  
c i  
a n  
l g  
a m  
n a  
d n  
r u  
e f  
s a  
u c  
lt t  
s u  
- ri  
o n  
ri g  
e o  
n u  
t t  
e c  
d o  
a m  
p e  
p s.  
r T  
o h  
a is  
c p  
h a  
f p  
o e

r c  
e e  
x u  
a ti  
m c  
i a  
n l  
e p  
s r  
t o  
h d  
e u  
i c  
m ti  
p o  
l n  
e i  
m n  
e J  
n o  
t r  
a d  
ti a  
o n  
n ,  
o f  
f o  
S c  
i u  
x si  
S n  
i g  
g o  
m n  
a t  
i h  
n e  
t t  
h a  
e b  
c l  
o e  
n t  
t c  
e o  
x m  
t p  
o r  
f e  
p s  
h si  
a o  
r n  
m st  
a a



---

g n  
e t  
f w  
o h  
r e  
A r  
M e  
O v  
F a  
6 ri  
2 a  
5 b  
t il  
a it  
b y  
l c  
e a  
t n  
s. si  
T g  
h n  
i i  
s fi  
p c  
h a  
a n  
s tl  
e y  
h a  
a ff  
s e  
b c  
e t  
e p  
n r  
i o  
d d  
e u  
n c  
ti t  
fi q  
e u  
d a  
a li  
s t  
a y  
c .  
ri T  
ti h  
c e  
a o  
l b  
p j  
o e  
i c

ti c  
v i  
e e  
i s  
s a  
t n  
o d  
a m  
p o  
p v  
l e

---

y  
t  
h  
e  
D  
M  
A  
I  
C

<sup>1</sup> Industrial Engineering Department, School of Engineering.  
The University of Jordan, Amman, Jordan  
<sup>2</sup> Engineering Department, Luminus Technical University  
College (LTUC), Amman, Jordan

Corresponding author:  
Mohammad D AL-Tahat, Industrial Engineering Department, School of  
Engineering. The University of Jordan, Queen Rania Street, Amman  
11942, Jordan.  
Email: [altahat@ju.edu.jo](mailto:altahat@ju.edu.jo)

m closer  
e to a  
t Six  
h Sigma  
o level  
d of  
o perfor  
l mance  
o ,  
g where  
y defects  
t are  
o limited  
r to  
e fewer  
d than  
u 3.4 per  
c millio  
e n  
s opport  
u unities  
c .  
h Wo  
i rking  
n in  
c partner  
o ship  
n with a  
s major  
i pharm  
s aceutic  
t al  
e firm,  
n

---

the and  
 research targete  
 h d  
 addres emplo  
 ses yee  
 key trainin  
 source g, are  
 s of introd  
 ineffici uced  
 ency to  
 in the stabili  
 produc ze  
 tion operati  
 line. ons.  
 The To  
 project mainta  
 involv in  
 es these  
 identif gains,  
 ying the  
 recurri study  
 ng also  
 issues, incorp  
 impro orates  
 ving ongoin  
 measu g  
 rement control  
 accura measu  
 cy, and res,  
 applyi includi  
 ng ng  
 analyti realtim  
 cal e  
 tools monito  
 to ring  
 uncov and  
 er the use  
 underl of  
 ying statisti  
 causes cal  
 of proces  
 variati s  
 on. control  
 Practic .  
 al Fin  
 change dings  
 s, such sugges  
 as t that  
 proces adopti  
 s ng Six  
 autom Sigma  
 ation, not  
 refined only  
 schedu reduce  
 ling, s

defects how  
 but Six  
 also Sigma  
 support can  
 its a serve  
 broader as a  
 its founda  
 culture tion  
 of for  
 impro buildin  
 vemen g more  
 t and resilie  
 accoun nt and  
 tability consist  
 . By ent  
 applyi produc  
 ng tion  
 structu enviro  
 red nment  
 metho s in a  
 ds compe  
 across titive  
 all industr  
 stages y. The  
 — paper  
 from details  
 diagno the  
 sis to practic  
 control al  
 —the applic  
 initiati ation  
 ve of  
 contrib DMAI  
 utes C to  
 practic the  
 al AMO  
 insight F625  
 s for tablet  
 pharm compr  
 aceutic ession  
 al proces  
 firms s,  
 seekin suppor  
 g to ted by  
 strengt real  
 hen produc  
 both tion  
 perfor data  
 mance and  
 and statisti  
 quality cal  
 . It analysi  
 also s.  
 illustra Each  
 tes phase

of the Obj  
metho ecti  
dology  
— ves  
from and  
route res  
cause  
identif ear  
ication ch  
to que  
control stio  
imple  
mentat ns  
ion— This  
is researc  
exami h aims  
ned to  
with elevate  
measu the  
rable efficie  
outco ncy  
mes. and  
The quality  
study standa  
conclu rds of  
des pharm  
with a aceutic  
discus al  
sion of manuf  
results acturin  
,  
validat g  
ion, proces  
and ses in  
implic Jordan  
ations ,  
for specifi  
wider cally  
use in targeti  
pharm ng the  
aceutic tablet  
al compr  
manuf ession  
acturin phase  
g. for the  
medic  
ation  
AMO  
F625,  
where  
variabi  
lity  
poses  
signifi  
cant

challenges, the  
 researches.  
 By highlighting  
 employment seeks  
 aiming to  
 the validation  
 DMAIC the  
 C effectiveness  
 (Definition of  
 e, Six  
 Measurement Sigma  
 re, method  
 Analytical  
 ze, issues in  
 Improving drivin  
 ve, g  
 and operational  
 Control  
 l) impro  
 methodo  
 dology  
 within Address  
 the Six Sigma  
 Sigma issues  
 framework  
 work, ending  
 the nonco  
 research informi  
 h ng  
 endeavor units  
 ors to in the  
 achievement  
 e a Six Sigma  
 Sigma session  
 Quality proces  
 y s, the  
 Level study  
 ( $6\sigma$ ), engage  
 signify s a  
 ing divers  
 substance team  
 ntial to  
 enhance explor  
 ements e  
 . solution  
 Collaborations and  
 orating optimi  
 with a ze  
 leading proces  
 g s  
 pharmaceutical  
 efficacy.  
 al firm Ultima  
 in tely,  
 Jordan this

---

research dology  
 h aims for  
 to proces  
 contrib s  
 ute enhanc  
 valuab ement  
 le that  
 insight encom  
 s to passes  
 the three  
 pharm distinc  
 aceutic t  
 al interpr  
 manuf etation  
 acturin s. As a  
 g statisti  
 sector, cal  
 advanc instru  
 ing the ment,  
 overar it is  
 ching concer  
 goals ned  
 of with  
 quality sustain  
 enhanc ing a  
 ement defect  
 and rate of  
 produc 3.4  
 tivity errors  
 throug per  
 h millio  
 system n  
 atic opport  
 impro unities  
 vemen . As a  
 t proced  
 initiati ural  
 ves. impro  
 vemen  
 t  
**Lite** frame  
**rat** work,  
**ure** it  
**revi** adhere  
**ew** s to  
 the  
 Six DMAI  
 Sigma C  
 constit (Defin  
 utes a e,  
 highly Measu  
 esteem re,  
 ed Analy  
 metho ze,  
 Impro

---

ve, ook1  
 and and  
 Contro Georg  
 l) e et  
 metho al.'s  
 dology The  
 , fat Six  
 which Sigma  
 acts as Pocket  
 its Toolbo  
 funda ok,2  
 mental offers  
 princip extensi  
 le. As ve  
 a directi  
 philos ves  
 ophica and  
 l pragm  
 stance, atic  
 Six instru  
 Sigma ments  
 unders for its  
 cores applic  
 the ation  
 necess across  
 ity of variou  
 genera s  
 ting sectors  
 produc .  
 ts and Critica  
 service l  
 s in succes  
 the s  
 most determ  
 efficie inants  
 nt and articul  
 effecti ated  
 ve by  
 manne Mehra  
 r bi3  
 possibl encom  
 e. pass  
 Found the  
 ational involv  
 literatu ement  
 re, of  
 includi senior  
 ng manag  
 Pyzde ement,  
 k and organi  
 Keller' zation  
 s The al  
 Six dedica  
 Sigma tion,  
 Handb profici



---

ent operati  
 project onal  
 manag efficie  
 ement, ncy  
 cultura and  
 l enviro  
 transfo nment  
 rmatio al  
 n, and perfor  
 ongoin mance  
 g ,  
 trainin particu  
 g, all larly  
 of within  
 which highly  
 are regulat  
 essenti ed  
 al for sectors  
 the such  
 sustain as  
 able pharm  
 imple aceutic  
 mentat als.  
 ion of Numer  
 Six ous  
 Sigma schola  
 initiati rly  
 ves. referen  
 Cherra ces  
 fi et have  
 al.4 deline  
 empha ated  
 sized lean as  
 the a  
 synerg system  
 istic ;  
 integra Kraje  
 tion of wski5  
 Six charac  
 Sigma terized  
 with a lean  
 lean system  
 manuf as  
 acturin “Oper  
 g and ations  
 sustain system  
 ability s that  
 practic maxim  
 es, ize the  
 illustra value  
 ting its added  
 impact by  
 on each  
 enhanc of a  
 ing compa

ny's closer  
activities to  
es by providing  
removing  
ng customers  
source of with  
waste exact  
and y what  
delays they  
from want.”  
them.” Building  
Woman on  
ck and this  
Jones, founder  
in tional  
their concep  
seminar,  
l work divers  
“The e  
Machi studies  
ne have  
That exami  
Changed  
ed the practice  
World, al  
”6 applic  
definitions  
d lean of lean  
as princip  
“doing less to  
more optimi  
and ze  
more produc  
with tion  
less proces  
and ses.  
less— One  
less investi  
human gation  
effort, utilize  
less d  
equip value  
ment, stream  
less mappi  
time, ng to  
and enhanc  
less e  
space workfl  
— ow  
while and  
comin eradic  
g ate  
closer ineffic  
and iencies

---

, optimi  
 culmin zing  
 ating Work  
 in in  
 impro Progre  
 ved ss  
 overall (WIP)  
 produc within  
 tivity.<sup>7</sup> Kanba  
 In n-  
 anothe govern  
 r ed  
 study, produc  
 the tion  
 amalg lines,  
 amatio alongs  
 n of a idethe  
 Just- progre  
 InTim ssivee  
 e (JIT) volutio  
 produc nofcon  
 tion tinuou  
 system simpro  
 with vemen  
 Kanba t  
 n and metho  
 sampli dologi  
 ng es,  
 inspect particu  
 ion larly  
 was throug  
 exami h the  
 ned, integra  
 demon tion of  
 stratin Lean  
 g its Six  
 efficac Sigma.  
 y in 9,10  
 waste Both  
 reducti lean  
 on and and  
 quality Six  
 control Sigma  
 enhanc repres  
 ement. ent  
 8 metho  
 Furthe dologi  
 rmore, es  
 researc aimed  
 h has at  
 also augme  
 explor nting  
 ed busine  
 strateg ss  
 ies for proces

s execut  
 perfor ing  
 mance tasks  
 . Six correct  
 Sigma, ly,  
 as a effecti  
 philos veness  
 ophica relates  
 l to  
 approa execut  
 ch, ing the  
 concen approp  
 trates riate  
 on tasks.  
 optimi Conse  
 zing quentl  
 the y,  
 quality Lean  
 of predo  
 produc minant  
 ts and ly  
 service aims  
 s at  
 throug efficie  
 h ncy,  
 defect Six  
 reducti Sigma  
 on and priorit  
 precisi zes  
 on quality  
 enhanc and  
 ement. effecti  
 In veness  
 contra , and  
 st, lean Lean  
 endeav Six  
 ors to Sigma  
 bolster amalg  
 produc amates  
 tion both to  
 efficie optimi  
 ncy by ze  
 eradic overall  
 ating perfor  
 waste mance  
 in all .  
 its T  
 manife he  
 station util  
 s. izat  
 While ion  
 efficie of  
 ncy Le  
 pertain an  
 s to Six

---

Sig nt  
 ma im  
 me por  
 tho tan  
 dol ce.  
 ogi As  
 es del  
 has ine  
 ass ate  
 um d  
 ed by  
 an Pa  
 inc nd  
 rea ya  
 sin et  
 gly al.<sup>1</sup>  
 piv <sup>1</sup>  
 ota the  
 l esc  
 rol alat  
 e ing  
 wit co  
 hin mp  
 the etit  
 pha ive  
 rm dy  
 ace na  
 uti mi  
 cal cs  
 sec wit  
 tor, hin  
 wh the  
 ere ma  
 rig rke  
 oro tpl  
 us ace  
 qua hav  
 lity e  
 ass co  
 ura mp  
 nce elle  
 and d  
 enh nu  
 anc me  
 em rou  
 ent s  
 s in pha  
 pro rm  
 ces ace  
 ses uti  
 are cal  
 of ent  
 par erp  
 am rise  
 ou s to

---

int act  
egr uri  
ate ng  
Le has  
an un  
Six der  
Sig go  
ma ne  
as sub  
a sta  
fun nti  
da al  
me exp  
nta ans  
l ion  
stra ,  
teg un  
y der  
for sco  
the rin  
adv g  
anc its  
em con  
ent seq  
of uen  
the tial  
ir im  
pro pac  
ces t  
ses. on  
Th bot  
e h  
cor pro  
pus ces  
of s  
lite effi  
rat cie  
ure ncy  
add and  
res pro  
sin duc  
g t  
Le qua  
an lity  
Six .  
Sig Al  
ma hur  
in ais  
pha h et  
rm al.<sup>1</sup>  
ace <sup>2</sup>  
uti acc  
cal ent  
ma uat  
nuf ed

---

the stre  
 con am  
 trib ma  
 uti ppi  
 ons ng  
 of for  
 Le the  
 an opt  
 Six imi  
 Sig zati  
 ma on  
 to of  
 the pro  
 ope ces  
 rati ses  
 ona and  
 l the  
 and Ma  
 fina lm  
 nci qui  
 al st  
 out ind  
 co ex  
 me for  
 s of the  
 pha eva  
 rm lua  
 ace tio  
 uti n  
 cal of  
 ma pro  
 nuf duc  
 act tivi  
 uri ty,  
 ng. bot  
 Th h  
 eir of  
 inv wh  
 esti ich  
 gat hav  
 ion e  
 enc furt  
 om her  
 pas aug  
 sed me  
 me nte  
 tho d  
 dol pro  
 ogi duc  
 es tio  
 suc n  
 h effi  
 as cac  
 val y.  
 ue AL

---

- par  
Tah tic  
at ula  
and rly  
Mo con  
ha cer  
m nin  
me g  
d<sup>13</sup> stra  
hav teg  
e ies  
exa for  
mi wa  
ned ste  
lea mi  
n- ni  
ori mi  
ent zati  
ed on  
des and  
ign effi  
acr cie  
oss ncy  
div enh  
ers anc  
e em  
ind ent.  
ust A  
ries my  
, ria  
pro d  
ffer of  
ing me  
val tho  
uab dol  
le ogi  
insi es  
ght has  
s bee  
tha n  
t scr  
are uti  
app niz  
lica ed  
ble as  
to pha  
the rm  
pha ace  
rm uti  
ace cal  
uti fir  
cal ms  
do con  
ma fro  
in, nt



---

|      |      |
|------|------|
| esc  | ind  |
| alat | ust  |
| ing  | ry.  |
| reg  | Th   |
| ula  | ese  |
| tor  | inn  |
| y    | ova  |
| obl  | tio  |
| iga  | ns   |
| tio  | bol  |
| ns   | ster |
| and  | the  |
| ma   | fun  |
| rke  | da   |
| t    | me   |
| pre  | nta  |
| ssu  | l    |
| res, | ai   |
| wit  | m    |
| h    | of   |
| the  | Six  |
| obj  | Sig  |
| ecti | ma,  |
| ve   | wh   |
| of   | ich  |
| im   | is   |
| pro  | to   |
| vin  | di   |
| g    | mi   |
| effi | nis  |
| cie  | h    |
| ncy  | var  |
| and  | iab  |
| qua  | ilit |
| lity | y    |
| ,    | wh   |
| the  | ile  |
| reb  | fos  |
| y    | teri |
| fac  | ng   |
| ilit | con  |
| ati  | tin  |
| ng   | uo   |
| sig  | us   |
| nifi | im   |
| can  | pro  |
| t    | ve   |
| adv  | me   |
| anc  | nt   |
| em   | and  |
| ent  | ope  |
| s    | rati |
| wit  | ona  |
| hin  | l    |
| the  | exc  |

---

elle pro  
nce duc  
in tio  
the n  
pha pro  
rm ces  
ace ses.  
uti Th  
cal ese  
rea dev  
lm. elo  
Su pm  
ch ent  
me s  
tho har  
dol nes  
ogi s  
es dat  
of a  
con ana  
tin lyti  
uo cs,  
us ma  
im chi  
pro ne  
ve lea  
me rni  
nt ng,  
hav and  
e rea  
bee l-  
n tim  
pro e  
pel mo  
led nit  
by ori  
tec ng  
hn to  
olo refi  
gic ne  
al Six  
adv Sig  
anc ma  
em im  
ent ple  
s me  
and nta  
the tio  
gro ns,  
wi the  
ng reb  
intr y  
ica ena  
cy bli  
of ng

---

mo of  
 re Six  
 pre Sig  
 cis ma,  
 e un  
 reg vei  
 ula lin  
 tio g  
 n no  
 of vel  
 pro pro  
 ces spe  
 s cts  
 var for  
 iab opt  
 les imi  
 and zati  
 enh on  
 anc and  
 ing enh  
 the anc  
 con em  
 sist ent  
 enc s in  
 y effi  
 of cie  
 pro ncy  
 duc .  
 ts. S  
 Th om  
 e e  
 am insi  
 alg ght  
 am s  
 ati fro  
 on m  
 of rec  
 Ind ent  
 ust stu  
 ry die  
 4.0 s  
 par on  
 adi lev  
 gm era  
 s gin  
 has g  
 furt Le  
 her an  
 bro Six  
 ade Sig  
 ned ma  
 the and  
 hor em  
 izo erg  
 ns ing

---

tec val  
 hn uab  
 olo le  
 gie insi  
 s to ght  
 enh s  
 anc int  
 e o  
 pha pro  
 rm duc  
 ace tivi  
 uti ty  
 cal traj  
 ma ect  
 nuf ori  
 act es  
 uri and  
 ng fac  
 ent tor  
 ails s  
 sev infl  
 era uen  
 l cin  
 crit g  
 ical effi  
 co cie  
 mp ncy  
 one <sup>14</sup>  
 nts: for  
 the ass  
 im ess  
 ple ing  
 me the  
 nta ope  
 tio rati  
 n ons  
 of wit  
 wi hin  
 nd the  
 ow pha  
 ana rm  
 lysi ace  
 s uti  
 and cal  
 the sec  
 Ma tor.  
 lm Th  
 qui e  
 st adv  
 ind anc  
 ex, em  
 wh ent  
 ich of  
 yie an  
 ld ord

---

ina d  
 l wit  
 log h  
 isti tab  
 c leti  
 reg ng  
 res app  
 sio ara  
 n tus,  
 mo the  
 del reb  
 for y  
 Fai con  
 lur trib  
 e uti  
 Mo ng  
 de to  
 and im  
 Eff pro  
 ect ve  
 s me  
 An nts  
 aly in  
 sis bot  
 (F h  
 M reli  
 EA abi  
 ) lity  
 has and  
 sig qua  
 nifi lity  
 can .<sup>15</sup>  
 tly Th  
 enh e  
 anc exa  
 ed mi  
 the nat  
 ide ion  
 nti of  
 fica the  
 tio im  
 n pac  
 and t of  
 pri dig  
 orit ital  
 izat tec  
 ion hn  
 of olo  
 fail gie  
 ure s  
 mo and  
 des Ind  
 ass ust  
 oci ry  
 ate 4.0

---

on izat  
ind ion  
ust of  
rial dat  
agi a  
lity ana  
has lyti  
pro cs  
vid in  
ed the  
me ma  
ani int  
ngf ena  
ul nce  
insi of  
ght me  
s dic  
int al  
o equ  
the ip  
pro me  
spe nt  
cti has  
ve illu  
evo stra  
luti ted  
on ho  
of w  
pha suc  
rm h  
ace ana  
uti lyti  
cal cs  
ma can  
nuf be  
act eff  
uri ecti  
ng vel  
in y  
ali har  
gn nes  
me sed  
nt to  
wit opt  
h imi  
Six ze  
Sig pro  
ma ces  
pri s  
nci reli  
ple abi  
s.<sup>16</sup> lity  
Th and  
e effi  
util cie

---

|                  |                  |
|------------------|------------------|
| ncy              | em               |
| wit              | erg              |
| hin              | enc              |
| pha              | y                |
| rm               | dep              |
| ace              | art              |
| uti              | me               |
| cal              | nt               |
| ma               | has              |
| nuf              | un               |
| act              | der              |
| uri              | sco              |
| ng               | red              |
| con              | the              |
| tex              | ver              |
| ts. <sup>1</sup> | sati             |
| <sup>7</sup>     | lity             |
| Th               | of               |
| e                | Six              |
| am               | Sig              |
| alg              | ma               |
| am               | too              |
| ati              | ls               |
| on               | acr              |
| of               | oss              |
| Six              | div              |
| Sig              | ers              |
| ma               | e                |
| wit              | do               |
| h                | ma               |
| sim              | ins,             |
| ula              | inc              |
| tio              | lud              |
| n                | ing              |
| mo               | hea              |
| del              | lth              |
| ing              | car              |
| in               | e                |
| an               | and              |
| eff              | pha              |
| ort              | rm               |
| to               | ace              |
| enh              | uti              |
| anc              | cal              |
| e                | s. <sup>18</sup> |
| effi             | By               |
| cie              | rne              |
| ncy              | et               |
| in a             | al.              |
| ped              | star             |
| iatr             | ted              |
| ic               | a                |
| hos              | co               |
| pit              | mp               |
| al               | reh              |

---

ens tab  
ive lets  
inv tha  
esti t  
gat inc  
ion orp  
int ora  
o te  
the ace  
app ta  
lica mi  
tio no  
n phe  
of n.  
a Th  
LS eir  
S res  
me ear  
tho ch  
dol em  
og plo  
y yed  
wit a  
hin co  
the mp  
pha reh  
rm ens  
ace ive  
uti LS  
cal S  
sec fra  
tor, me  
[19](#) wo  
wit rk  
h tha  
par t  
tic sys  
ula te  
r ma  
em tica  
pha lly  
sis ide  
on nti  
the fied  
ma ,  
nuf cla  
act ssi  
uri fied  
ng ,  
pro and  
ces eli  
s of mi  
ana nat  
lge ed  
sic ine



---

|      |      |
|------|------|
| ffic | sav  |
| ien  | ing  |
| cie  | s    |
| s    | nea  |
| thr  | rin  |
| ou   | g    |
| gh   | hal  |
| a    | f a  |
| cus  | mil  |
| to   | lio  |
| mi   | n    |
| zed  | dol  |
| sev  | lars |
| en-  | .    |
| ste  | Th   |
| p    | e    |
| pro  | fin  |
| ble  | din  |
| m-   | gs   |
| sol  | of   |
| vin  | this |
| g    | res  |
| pro  | ear  |
| toc  | ch   |
| ol.  | hig  |
| Thi  | hli  |
| s    | ght  |
| me   | the  |
| tho  | piv  |
| dic  | ota  |
| al   | l    |
| app  | fun  |
| roa  | cti  |
| ch   | on   |
| suc  | of   |
| ces  | Le   |
| sfu  | an   |
| lly  | Six  |
| add  | Sig  |
| res  | ma   |
| sed  | in   |
| the  | aug  |
| iss  | me   |
| ue   | nti  |
| at   | ng   |
| han  | pha  |
| d    | rm   |
| and  | ace  |
| cul  | uti  |
| mi   | cal  |
| nat  | pro  |
| ed   | duc  |
| in   | tio  |
| cos  | n    |
| t    | by   |

---

red im  
uci ple  
ng me  
wa nta  
ste, tio  
enh n  
anc of  
ing LS  
effi S  
cie wit  
ncy hin  
, the  
and pha  
opt rm  
imi ace  
zin uti  
g cal  
ove sec  
rall tor,  
ope [20](#)  
rati wh  
ona ere  
l the  
per y  
for eva  
ma lua  
nce ted  
. the  
Bel app  
l et rop  
al. riat  
con ene  
duc ss  
ted of  
an LS  
ana S  
lysi wit  
s hin  
reg a  
ard me  
ing tic  
the ulo  
eth usl  
ical y  
im reg  
pli ula  
cati ted  
ons fra  
ass me  
oci wo  
ate rk.  
d Th  
wit e  
h inv  
the esti

---

gat ing  
 ion pro  
 eva ces  
 lua s  
 ted effi  
 pot cie  
 ent ncy  
 ial wit  
 eth h  
 ical eth  
 dil ical  
 em acc  
 ma ou  
 s nta  
 ass bili  
 oci ty  
 ate to  
 d up  
 wit hol  
 h d  
 Six co  
 Sig mp  
 ma lia  
 thr nce  
 ou and  
 gh int  
 the egr  
 len ity  
 s of in  
 var pha  
 iou rm  
 s ace  
 eth uti  
 ical cal  
 fra ope  
 me rati  
 wo ons  
 rks. .  
 Th Col  
 eir lect  
 res ive  
 ear ly,  
 ch the  
 acc se  
 ent stu  
 uat die  
 es s  
 the un  
 nec der  
 ess sco  
 ity re  
 of the  
 har ess  
 mo ent  
 niz ial

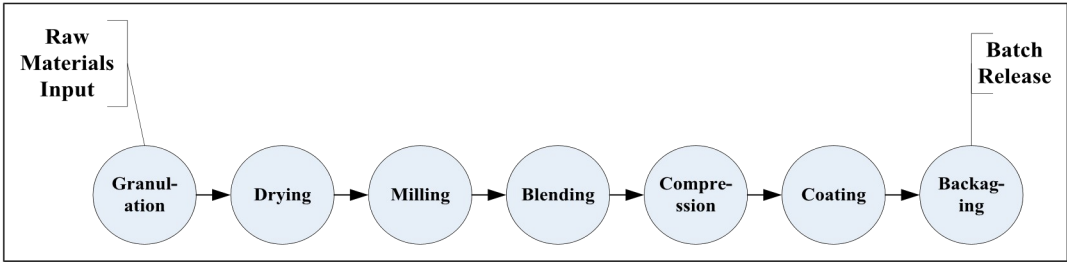
---

rol nuf  
e act  
of uri  
Le ng  
an pro  
Six ces  
Sig ses.  
ma The  
in overar  
fos ching  
teri theme  
ng concer  
con ning  
tin the  
uo imple  
us mentat  
im ion of  
pro LSS in  
ve this  
me discou  
nt, rse  
lev pertain  
era s to its  
gin functio  
g n in  
dat enhanc  
a- ing  
inf proces  
or s  
me efficie  
d ncy,  
insi produc  
ght t  
s, quality  
and , and  
int operati  
egr onal  
ati perfor  
ng mance  
em via a  
erg synthe  
ing sis of  
tec structu  
hn red  
olo metho  
gie dologi  
s to es,  
refi datace  
ne ntric  
pha decisio  
rm n-  
ace makin  
uti g, and  
cal strateg  
ma ies

aimed  
at  
contin  
uous  
impro  
vement.  
T  
h  
e  
t  
a  
b  
l  
e  
t  
  
p  
r  
o  
d  
u  
c  
t  
i  
o  
n  
  
p  
r  
o  
c  
e  
s  
s  
  
i  
n  
  
p  
h  
a  
r  
m  
a  
c  
e

u  
t  
i  
c  
a  
l  
s  
The  
produc  
tion is  
a  
multist  
age  
proces  
s  
during  
which  
raw  
materi  
als  
underg  
o  
several  
change  
s  
in  
their  
physic  
al  
charac  
teristic  
s  
before  
the  
final  
tablet  
is  
produc  
ed.  
The  
flowch  
art  
illustra  
ting  
the  
tablet  
produc  
tion  
proces  
ses is  
shown  
in

Figure  
1.  
The  
tablet  
produc  
tion  
proces  
ses  
includ  
e  
granul  
ation,  
drying  
,  
millin  
g,  
blendi  
ng,  
compr  
ession,  
coatin  
g, and  
tablet  
packag  
ing.  
The  
raw  
materi  
als  
used  
depend  
on  
the  
Figure  
1.  
Tablet  
produc  
tion  
sequen  
ce of  
process  
es.



Granulation is the process of collecting powder particles together with adhesive, forming a bridge between particles in a powder granulator to increase tensile strength and create the required granules. Wet granulation and dry granulation are the two types of granulation stages used in pharmaceutical industries. Wet Granulation, in this stage, granules are formed by adding granulation liquid to the powder particles in a high speed granulator to produce wet granules. The granulation liquid contained a solvent that must be removed by drying. Dry Granulation, granules are formed in this

stage Dry  
 without ing:  
 t using Drying  
 a is the  
 liquid proces  
 solutio s of  
 n, as evapor  
 some ating  
 granul the  
 es may liquid  
 be contai  
 sensiti ned  
 ve to within  
 moistu aggreg  
 re and ates  
 heat. produc  
 The ed by  
 primar the  
 y wet  
 powde granul  
 r ation  
 particl proces  
 es are s.  
 collect Mil  
 ed ling:  
 under In this  
 high stage,  
 pressu the  
 re unifor  
 using m  
 a roll particl  
 compa e size  
 ctor. of  
 Three ingredi  
 critical ents is  
 param reduce  
 eters d  
 affecti using  
 ng the a  
 granul hamm  
 ation er mill  
 proces machi  
 s are ne,  
 the which  
 amoun contai  
 t of ns a  
 liquid, series  
 mixing of  
 time, hamm  
 and ers to  
 granul reduce  
 ation particl  
 time. e sizes  
 by  
 impact



---

and Tab  
 provid let  
 e a Compr  
 specifi ession:  
 c This  
 particl stage  
 e size involv  
 for es the  
 easier initial  
 mixing elimin  
 . ation  
 Fin of  
 al voids  
 Blendi and  
 ng: the  
 Final rearran  
 blendi gemen  
 ng is t of  
 perfor particl  
 med es  
 using within  
 a the die  
 Doubl cavity.  
 e Cone The  
 Mixer. tablet  
 Powde compr  
 r is ession  
 mixed proces  
 first, s will  
 follow be  
 ed by explai  
 the ned  
 additio briefly  
 n of in the  
 lubrica next  
 nts, section  
 and .  
 the Coa  
 mixed ting:  
 powde Tablet  
 r is coatin  
 sifted g is  
 to the  
 attain proces  
 a s of  
 unifor coatin  
 mly g  
 blende compo  
 d nents  
 powde onto  
 r for the  
 compr tablets  
 ession. using  
 heated  
 air to

facilitate solving of  
 solvent active  
 ingredients  
 evaporation. and  
 Coating suitable  
 strength pharm  
 aceutical  
 and al  
 tough excipients.  
 The  
 tablet  
 and tablet  
 improvement compr  
 ves session  
 taste. machi

Packaging is a  
 high-speed  
 : speed  
 Packaging mecha  
 nical  
 device  
 that  
 protect compr  
 ing the esses  
 final the  
 product ingredi  
 ents  
 for into  
 storage the  
 e, the  
 distribution, require  
 d  
 and shape  
 use. with  
 high  
 precision

Tablet on  
 within  
 the die  
 cavity.  
 As  
 mentioned  
 previously,  
 the  
 type of  
 tablet  
 discussed in  
 this  
 paper  
 is

symbolized by AMO F625 tablet, specifications of these tablets will be illustrated later. The compression machine, including upper and lower punches. Figure 2 illustrates the working principles of the compression machine and other main components. The

Tablet  
et  
co  
mp  
ress  
ion  
pro  
ces  
s  
pha  
ses  
As  
illu  
stra  
ted  
in  
Fig  
ure  
3,  
the  
tabl  
et  
co  
mpr  
essi  
on  
pro  
ces  
s  
con  
sist  
s of  
fou  
r  
pha  
ses:  
filli  
ng,  
met  
erin  
g,  
co  
mpr  
essi  
on,  
and  
ejec  
tion  
.  
The  
se

phases are divided into two stages: the power stage and the compression stage. The machine's annual filling capacity. This stage involves manual filling of the compression chamber with the powder. The power

---

die the  
 s to met  
 ena erin  
 ble g  
 the ca  
 exa m.  
 ct C  
 ma om  
 ss pre  
 of ssio  
 po n  
 wd Sta  
 er ge:  
 to Tab  
 be lets  
 co are  
 mp pro  
 ress duc  
 ed ed  
 into by  
 tabl co  
 ets. mpr  
 The essi  
 exa ng  
 ct the  
 qua po  
 ntit wd  
 y of er  
 po in  
 wd the  
 er die  
 is cav  
 con ity  
 trol usi  
 led ng  
 by har  
 the den  
 hei ed  
 ght stee  
 of l  
 the pun  
 low che  
 er s.  
 pun The  
 ch forc  
 in e  
 the exe  
 die, rted  
 whi in  
 ch the  
 is dies  
 con is  
 trol pre  
 led cise  
 by ly

con arat  
 trol ely,  
 led and  
 to aut  
 ens o-  
 ure reje  
 eac cte  
 h d  
 tabl ins  
 et pec  
 rec tion  
 eiv syst  
 es ems  
 the test  
 sa the  
 me tabl  
 for ets  
 ce for  
 and co  
 pre mpl  
 ssu ian  
 re. ce  
 E wit  
 ject h  
 ion spe  
 Sta cifi  
 ge: cati  
 Tab ons  
 lets .  
 are T  
 ejec he  
 ted acc  
 fro ept  
 m abl  
 the e  
 die val  
 by ues  
 the of  
 low Crit  
 er ical  
 pun to  
 ch. Qu  
 No alit  
 n- y  
 con (CT  
 for Q)  
 min cha  
 g ract  
 tabl eris  
 ets tics  
 are for  
 coll the  
 ecte tabl  
 d ets  
 sep pro

duced by the compression machine are provided in Table 1. Definition of conformity and a non conformity unit No nco nfo rmi ty

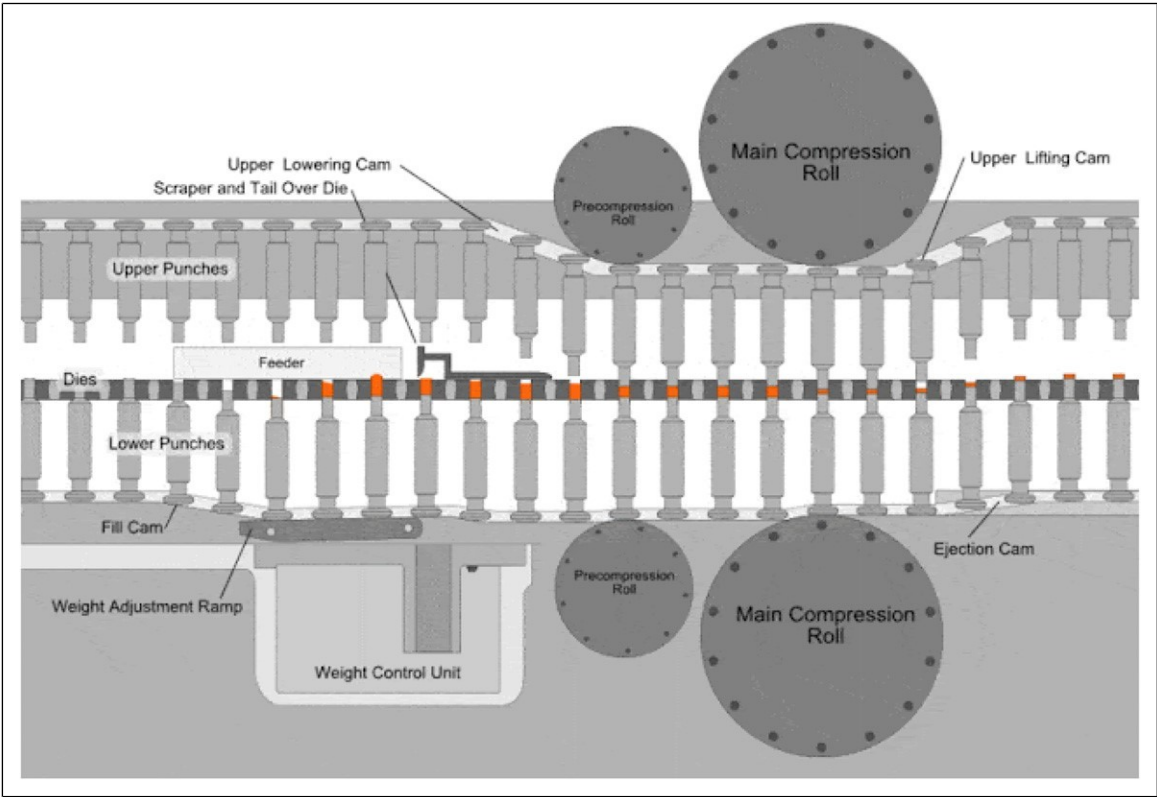


Figure 2. Tablet compression machine components and working principles.<sup>21</sup>

is deli

---

nea dne  
ted ss,  
as or  
the thic  
stat kne  
e ss,  
wh as  
erei spe  
n cifi  
an ed  
A in  
M [Tab](#)  
OF [le](#)  
625 [1. A](#)  
tabl non  
et con  
fail for  
s to min  
co g  
mp unit  
ly is  
wit defi  
h ned  
one as  
or any  
mu AM  
ltip OF  
le 625  
des tabl  
ign et  
ate that  
d doe  
crit s  
ical not  
-to- ful  
qua fill  
lity the  
par esta  
am blis  
eter hed  
s, crit  
whi eria  
ch for  
incl che  
ude mic  
che al  
mic co  
al mp  
co osit  
mp ion,  
osit wei  
ion, ght,  
wei har  
ght, dne  
har ss,



---

or d  
 thic e  
 kne v  
 ss i  
 as a  
 det t  
 aile e  
 d in s  
 Tab  
 le f  
 1. r  
 In o  
 part m  
 icul  
 ar, t  
 a h  
 tabl e  
 et  
 is s  
 clas t  
 sifi a  
 ed n  
 as d  
 non a  
 con r  
 for d  
 min s  
 g  
 if: s  
 e  
 t  
 (1) T  
 h  
 e f  
 o  
 c r  
 h t  
 e h  
 m  
 i b  
 c y  
 a  
 l t  
 h  
 c e  
 o  
 m U  
 p n  
 o i  
 s t  
 i e  
 t d  
 i  
 o S  
 n t  
 a

t l  
e l  
s i  
n  
P  
h (   
a i  
r n  
m  
a t  
c h  
o e  
p  
e f  
i o  
a r  
m  
(  
U o  
S f  
P  
) a  
. m  
(2) I o  
t x  
i  
c c  
o i  
n l  
t l  
a i  
i n  
n  
s t  
a r  
n i  
h  
y  
a d  
m r  
o a  
u t  
n e  
t )  
  
o t  
f h  
a  
a t  
m  
o i  
x s  
i  
c b  
i e

---

l o  
 o f  
 w c  
 5 l  
 0 a  
 0 v  
 u  
 m l  
 i a  
 l n  
 l i  
 i c  
 g  
 r a  
 a c  
 m i  
 s d  
 .  
 (3) I (   
 t a  
 s  
 i  
 n p  
 c o  
 l t  
 u a  
 d s  
 e s  
 s i  
 u  
 l m  
 e  
 s c  
 s l  
 a  
 t v  
 h u  
 a l  
 n a  
 n  
 1 a  
 2 t  
 5 e  
 )  
 m .  
 i (4) T  
 l h  
 l e  
 i  
 g w  
 r e  
 a i  
 m g  
 s h

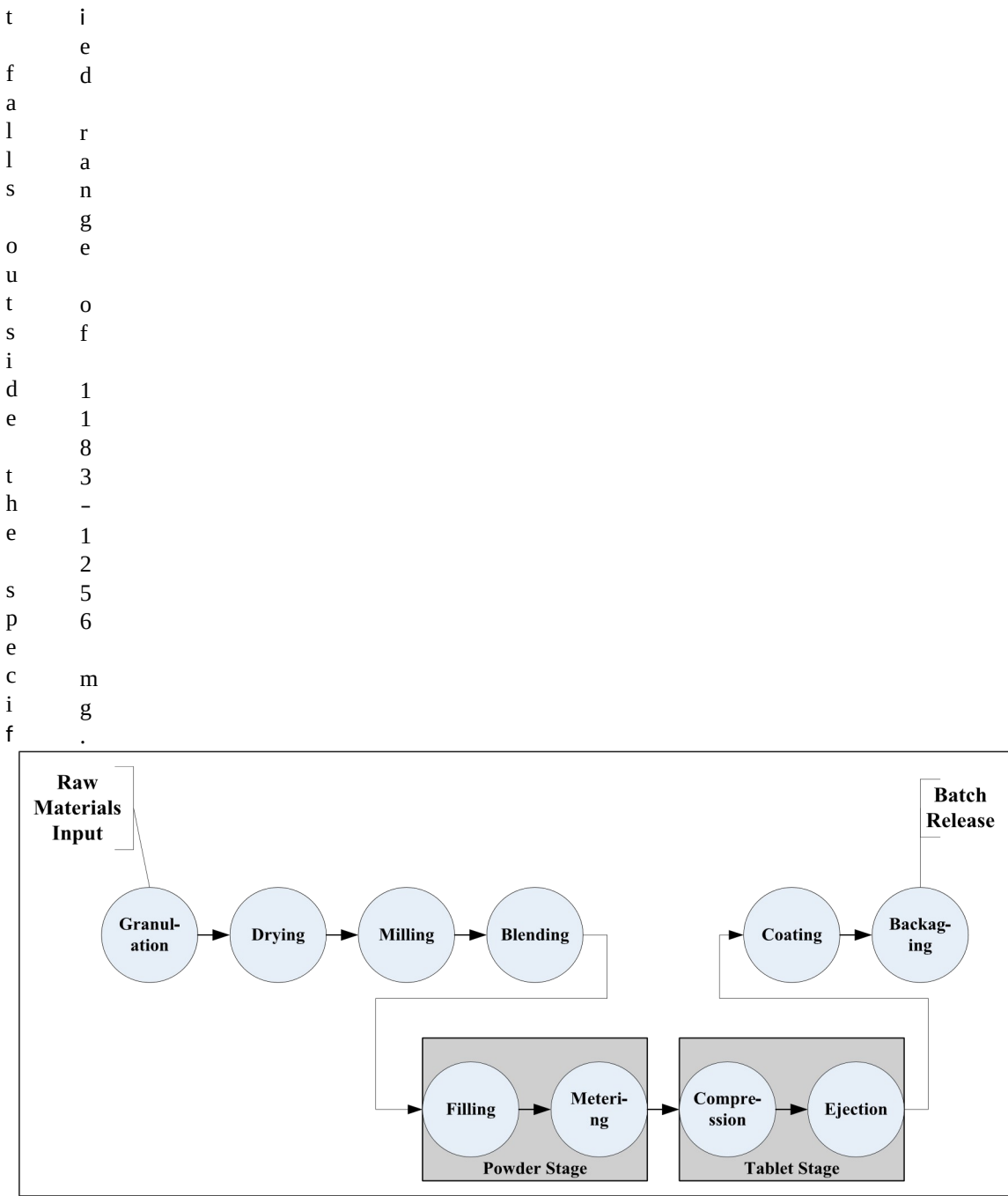


Figure 3. Tablet compression process phases.

Tablet compression process phases.

on  
s  
of  
A  
M  
OF  
62  
5  
ta  
ble  
ts.

| Critical to quality  | Acceptable values                                                                                                                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemical composition | According to the United States Pharmacopeia (USP) standards<br>- 500 milligrams of amoxicillin, the form of amoxicillin trihydrate<br>- 125 milligrams of clavulanic acid, in the form of potassium clavulanate |
| Weight               | 1183–1256 mg                                                                                                                                                                                                    |
| Hardness             | 130–395 N                                                                                                                                                                                                       |
| Thickness            | 7.50–8.25 mm                                                                                                                                                                                                    |

- (5) The ed  
har ran  
dne ge  
ss of  
doe 7.50  
s -  
not 8.25  
resi mm  
de .  
wit  
hin  
the D  
stip M  
ulat Al  
ed C  
ran  
ge ap  
of pr  
130 oa  
- ch  
395  
N. fra  
(6) The m  
thic ew  
kne or  
ss is k  
not  
cont The  
aine DMA  
d IC  
wit meth  
hin odolo  
the gy is  
pres a  
crib

---

struct ure, and Contr ol) appro ach as a syste matic frame work for conti nuous impro vement, with the goal of impr oving proce ss perfo rman ce. It is based on five steps: Defin e Meas ure, Anal yze, Impr ove, and Contr ol. This paper intro duces the DMA IC (Defi ne, Meas ure, Anal yze, Impr

how DMA IC impacts process design, performance, and capability in relation to Critical Quality (CTQ) parameters. The following sections explain in how each DMA IC step was implemented to achieve these objectives.

**De fin e**

The Defin e phase of the DMA IC approach presented challenges in selecting the suitable process and defining defect criteria. The tablet compression process was identified as suitable, focus ing on AMO F625 tablets.

**N on**

---

- ect  
co ch  
nf art  
or er  
mi de  
ng ve  
un lo  
its p  
we m  
re en  
de t,  
fin an  
ed d  
as pr  
de oc  
fe es  
cts s  
. m  
Ac ap  
tiv pi  
iti ng  
es .  
in Ro  
cl ot  
ud ca  
ed us  
Fo es  
cu id  
s en  
Gr tifi  
ou ed  
p m  
(F an  
G) ua  
dis l  
cu pr  
ssi oc  
on ed  
s, ur  
int es,  
er ne  
vi gl  
ew ect  
s, ed  
gr au  
ou to  
pt m  
hi ate  
nk d  
se to  
ssi ols  
on ,  
s, m  
pr ac  
oj hi



---

ne pr  
set oc  
tin ed  
g ur  
ch es,  
an sc  
ge he  
s, du  
lac lin  
k g  
of M  
m Os  
ai ,  
nt ad  
en he  
an rin  
ce, g  
an to  
d St  
dis an  
or da  
ga rd  
ni O  
ze pe  
d rat  
M in  
an g  
uf Pr  
act oc  
uri ed  
ng ur  
Or es  
de (S  
rs O  
( Ps  
M ),  
Os en  
) su  
Pr rin  
op g  
os lu  
ed bri  
sol ca  
uti nt  
on su  
s pp  
in ly,  
cl co  
ud nd  
ed uc  
au tin  
to g  
m m  
ati ai  
ng nt

---

en s  
an flo  
ce, w  
an ch  
d art  
pr an  
ov d  
idi Su  
ng pp  
tra lie  
ini r-  
ng In  
. pu  
Th tPr  
e oc  
pr es  
oj s-  
ect O  
ch ut  
art pu  
er t-  
ou Cu  
tli sto  
ne m  
d er  
th (S  
e IP  
pr O  
ob C)  
le di  
m, ag  
go ra  
al, m  
bu ai  
sin de  
es d  
s in  
ca un  
se, de  
tea rst  
m an  
sel di  
ect ng  
io th  
n, e  
an cu  
d rre  
pl nt  
an pr  
. oc  
Th es  
e s.  
pr SI  
oc P  
es O

---

C: po  
 Th te  
 e nti  
 SI al  
 P ro  
 O ot  
 C ca  
 (S us  
 up es  
 pli in  
 er- flu  
 In en  
 pu ci  
 t- ng  
 Pr pe  
 oc rfo  
 es rm  
 s- an  
 O ce  
 ut an  
 pu d  
 t- hi  
 Cu gh  
 sto lig  
 m hti  
 er) ng  
 di pr  
 ag ob  
 ra ab  
 m le  
 off so  
 ers ur  
 a ce  
 co s  
 m of  
 pr pe  
 eh rfo  
 en rm  
 siv an  
 e ce  
 de iss  
 pi ue  
 cti s.  
 on It  
 of als  
 th o  
 e su  
 pr gg  
 oc est  
 es s  
 s, w  
 re ha  
 ve t  
 ali as  
 ng pe

cts di  
ca ag  
n ra  
an m  
d de  
sh tai  
ou lin  
ld g  
be th  
m e  
ea ta  
su bl  
re et  
d, co  
inf m  
or pr  
mi es  
ng sio  
th n  
e pr  
da oc  
ta es  
co s.  
lle  
cti  
on M  
pl ea  
an sur  
in e  
th m  
e en  
M t  
ea  
su  
re  
ph  
as  
e.  
Re  
fer  
to  
[Fi](#)  
[gu](#)  
[re](#)  
[4](#)  
be  
lo  
w  
for  
th  
e  
SI  
P  
O  
C

In this phase, the DM AIC approach was meticulously applied to address the issue

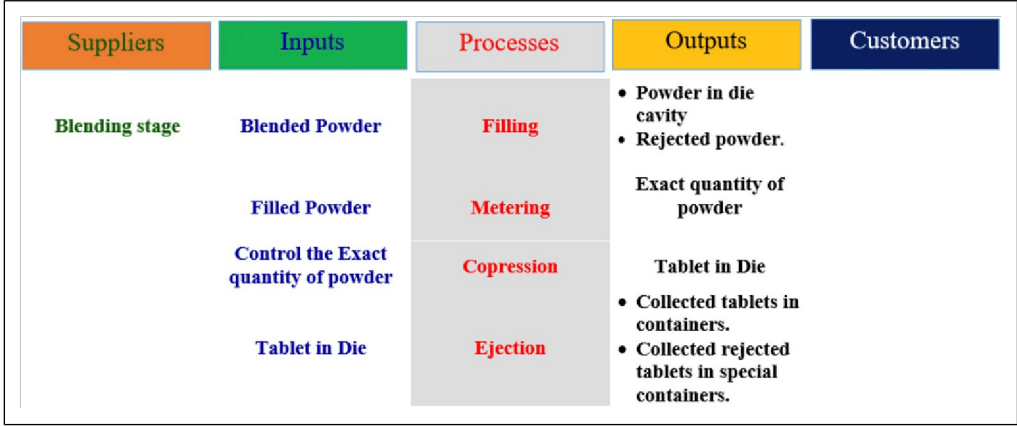


Figure 4. Tablet compression process SIPOC diagram.

With this rejection ion units in the tablet comp ressi on proce ss. To ensur e a repre senta tive samp le, a histor ical data

---

|     |     |
|-----|-----|
| od  | w   |
| uc  | as  |
| ti  | de  |
| on  | li  |
| co  | be  |
| nd  | ra  |
| iti | tel |
| on  | y   |
| s,  | se  |
| th  | le  |
| e   | ct  |
| st  | ed  |
| ud  | to  |
| y   | sp  |
| ut  | an  |
| ili | ro  |
| ze  | ut  |
| d   | in  |
| da  | e   |
| ta  | op  |
| fr  | er  |
| o   | ati |
| m   | on  |
| te  | al  |
| n   | fl  |
| pr  | uc  |
| od  | tu  |
| uc  | ati |
| ti  | on  |
| on  | s,  |
| ba  | in  |
| tc  | cl  |
| he  | ud  |
| s   | in  |
| co  | g   |
| lle | ch  |
| ct  | an  |
| ed  | ge  |
| ov  | s   |
| er  | in  |
| an  | sh  |
| 8-  | ift |
| m   | sc  |
| on  | he  |
| th  | du  |
| pe  | le  |
| ri  | s,  |
| od  | eq  |
| .   | ui  |
| T   | p   |
| hi  | m   |
| s   | en  |
| ra  | t   |
| ng  | m   |
| e   | ai  |

---

nt e  
en pr  
an oc  
ce es  
, s  
an un  
d de  
en r a  
vi ra  
ro ng  
n e  
m of  
en ty  
tal pi  
co ca  
nd l  
iti ci  
on rc  
s. u  
T m  
he st  
ch an  
oi ce  
ce s,  
w pr  
as ov  
no id  
t in  
ar g  
bi a  
tr re  
ar ali  
y; sti  
ra c  
th ba  
er, si  
it s  
w fo  
as r  
in as  
fo se  
r ss  
m m  
ed en  
by t.  
th Pr  
e io  
ne r  
ed to  
to an  
ob al  
se ys  
rv is,  
e th  
th e

---

|     |     |
|-----|-----|
| da  | e   |
| ta  | ne  |
| w   | ce  |
| er  | ss  |
| e   | ar  |
| te  | y   |
| st  | st  |
| ed  | ati |
| fo  | sti |
| r   | ca  |
| no  | l   |
| r   | as  |
| m   | su  |
| ali | m   |
| ty, | pt  |
| wi  | io  |
| th  | ns  |
| re  | .   |
| su  | A   |
| lts | dd  |
| in  | iti |
| di  | on  |
| ca  | all |
| ti  | y,  |
| ng  | m   |
| th  | ea  |
| at  | su  |
| th  | re  |
| e   | m   |
| di  | en  |
| str | t   |
| ib  | sy  |
| ut  | st  |
| io  | e   |
| ns  | m   |
| at  | ev  |
| bo  | al  |
| th  | ua  |
| th  | ti  |
| e   | on  |
| po  | us  |
| w   | in  |
| de  | g   |
| r   | G   |
| an  | au  |
| d   | ge  |
| ta  | R   |
| bl  | &   |
| et  | R   |
| st  | co  |
| ag  | nf  |
| es  | ir  |
| m   | m   |
| et  | ed  |
| th  | th  |



---

|     |     |
|-----|-----|
| at  | in  |
| m   | fo  |
| ea  | rc  |
| su  | ed  |
| re  | co  |
| m   | nf  |
| en  | id  |
| t-  | en  |
| re  | ce  |
| lat | in  |
| ed  | th  |
| va  | e   |
| ri  | re  |
| ati | lia |
| on  | bi  |
| w   | lit |
| as  | y   |
| m   | of  |
| in  | th  |
| i   | e   |
| m   | co  |
| al, | lle |
| ac  | ct  |
| co  | ed  |
| un  | da  |
| ti  | ta. |
| ng  | F   |
| fo  | ur  |
| r   | th  |
| le  | er  |
| ss  | va  |
| th  | li  |
| an  | da  |
| on  | ti  |
| e   | on  |
| pe  | ca  |
| rc  | m   |
| en  | e   |
| t   | fr  |
| of  | o   |
| to  | m   |
| tal | co  |
| ob  | nt  |
| se  | ro  |
| rv  | l   |
| ed  | ch  |
| va  | ar  |
| ri  | t   |
| ab  | an  |
| ili | al  |
| ty. | ys  |
| T   | is, |
| hi  | w   |
| s   | hi  |
| re  | ch  |

---

|     |     |
|-----|-----|
| sh  | ca  |
| o   | nt  |
| w   | va  |
| ed  | ri  |
| th  | ati |
| at  | on  |
| th  | be  |
| e   | tw  |
| pr  | ee  |
| oc  | n   |
| es  | ba  |
| s   | tc  |
| re  | he  |
| m   | s   |
| ai  | or  |
| ne  | op  |
| d   | er  |
| st  | at  |
| ab  | or  |
| le  | s.  |
| th  | Ta  |
| ro  | ke  |
| ug  | n   |
| ho  | to  |
| ut  | ge  |
| th  | th  |
| e   | er, |
| sa  | th  |
| m   | es  |
| pl  | e   |
| in  | fi  |
| g   | nd  |
| pe  | in  |
| ri  | gs  |
| od  | co  |
| .   | nf  |
| A   | ir  |
| N   | m   |
| O   | th  |
| V   | at  |
| A   | th  |
| re  | e   |
| su  | se  |
| lts | le  |
| al  | ct  |
| so  | ed  |
| su  | sa  |
| gg  | m   |
| es  | pl  |
| te  | e   |
| d   | of  |
| no  | fe  |
| si  | rs  |
| gn  | a   |
| ifi | ba  |

---

la m  
nc a  
ed fr  
an a  
d m  
cr e  
ed w  
ib or  
le k.  
fo Th  
un e data  
da inclu  
ti des  
on meas  
fo urem  
r ents  
ev of  
al nonc  
ua onfor  
ti ming  
ng units  
th (in  
e gram  
ef s) at  
fe both  
cti the  
ve powd  
ne er  
ss and  
of tablet  
th stage  
e s of  
i the  
m produ  
pr ction  
ov proce  
e ss for  
m AMO  
en F625.  
ts These  
in nonc  
tr onfor  
od ming  
uc units  
ed repre  
th sent  
ro defec  
ug t  
h rates  
th withi  
e n  
Si each  
x batch  
Si ,  
g whic

---

h are tion,  
 critic was  
 al calcul  
 indic ated  
 ators to  
 of start  
 proce a  
 ss baseli  
 perfo ne for  
 rman curre  
 ce. nt  
 Defe avera  
 ct ge  
 rate perfo  
 for rman  
 each ce  
 batch and  
 at varia  
 both bility,  
 the accor  
 powd dingl  
 er y  
 and gauge  
 tablet and  
 stage capab  
 s is ility  
 estim analy  
 ated sis of  
 by the  
 dividi proce  
 ng ss  
 the will  
 numb be  
 er of cond  
 nonc ucted  
 onfor to  
 ming identi  
 units fy  
 by trend  
 the s,  
 total patter  
 produ ns,  
 ced and  
 units. poten  
 Descr tial  
 iptive areas  
 statist for  
 ics, impro  
 inclu veme  
 ding nt.  
 mean Norm  
 and ality  
 stand test  
 ard for  
 devia data

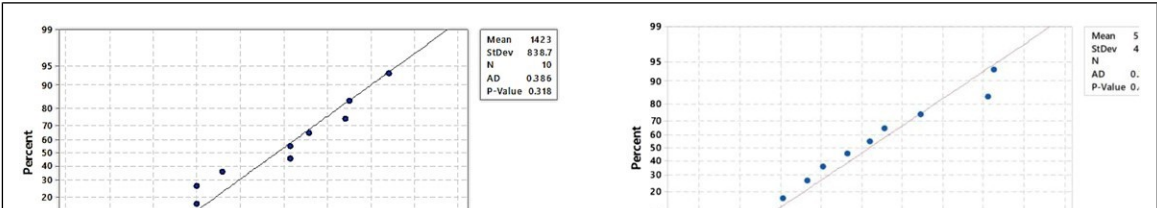
sample, repeatability, and reproducibility (R&R) analysis, and ANOVA analysis will be performed in the subsequent sections to

Analy-sis The observed variation in product quality characteristics can be categorized into two components:

Table 2. Data sample encompassing ten batches produced of AMOF625 over 8 months.

| Batch No.          | Produced units (gram) | Total number of nonconforming Units (gram) |
|--------------------|-----------------------|--------------------------------------------|
| Powder stage       |                       |                                            |
| 1                  | 1,17,600              | 1570                                       |
| 2                  | 1,317,600             | 1790                                       |
| 3                  | 1,317,600             | 2700                                       |
| 4                  | 1,317,600             | 360                                        |
| 5                  | 878,400               | 1570                                       |
| 6                  | 878,400               | 500                                        |
| 7                  | 1,317,600             | 790                                        |
| 8                  | 1,317,600             | 2250                                       |
| 9                  | 1,317,600             | 2200                                       |
| 10                 | 1,317,600             | 500                                        |
| Mean               | 1,229,760             | 1423                                       |
| Standard deviation | 185,183               | 839                                        |
| Tablet Stage       |                       |                                            |
| 1                  | 1,315,560             | 2610                                       |
| 2                  | 1,314,560             | 5455                                       |
| 3                  | 1,316,900             | 8595                                       |
| 4                  | 1,314,680             | 6370                                       |
| 5                  | 874,040               | 40                                         |
| 6                  | 879,060               | 12,780                                     |
| 7                  | 1,310,850             | 1600                                       |
| 8                  | 1,315,090             | 4080                                       |
| 9                  | 1,311,520             | 13,130                                     |
| 10                 | 1,317,150             | 130                                        |
| Mean               | 1,226,941             | 5479                                       |
| Standard deviation | 184,687               | 4782                                       |

variati  
ion  
and  
meas  
urem  
ent  
variati  
ion.  
The  
meas  
urem  
ent  
variati  
ion is  
furth  
er  
disse  
cted  
into  
R&R  
analy  
sis—  
repea  
tabili  
ty  
(R)  
and



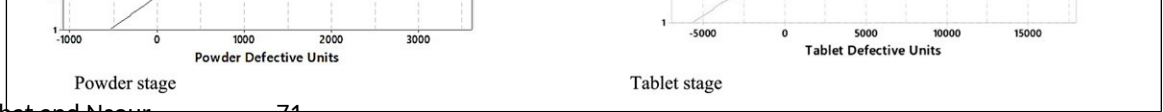


Figure 5. Normality test for data sample. e. reproducibility (R). The analysis aims to determine the extent to which the process variability is influenced by the measurement system's variability. Repeatability and reproducibility refer to two distinct concepts

points in measurement uncertainty analysis. Repeatability pertains to the consistency of measurements taken under the same conditions, while reproducibility refers to the consistency of measurements taken under different conditions. The analysis aims to determine the extent to which the process variability is influenced by the measurement system's variability. Repeatability and reproducibility refer to two distinct concepts

repro both  
ducib the  
ility powd  
conce er  
rns stage  
the and  
variat tablet  
ion stage  
obser  
ved  $\sigma_{\text{obs}}^2$   
in the  $\frac{1}{4}\sigma_{\text{part}}$   
avera  $s^2$   
ge of  $\sigma_{\text{mea}}$   
meas suremen  
urem  $t^2$   
ents Th  
( $\sigma_{\text{parts}}^2$  e  
) total  
cond varia  
ucted bility  
by can  
differ be  
ent divid  
appra ed as  
isers follo  
yet wing  
using equat  
the ion  
same below  
meas .  
uring Wher  
instru e  
ment  $\sigma_{\text{total}}^2$   
to is the  
asses total  
s varia  
identi nce,  
cal  $\sigma_{\text{system}}$   
chara  $^2$  is  
cteris the  
tics meas  
on urem  
the ent  
same syste  
part.<sup>22</sup> m  
Belo varia  
w are nce,  
the  $\sigma_{\text{repeata}}$   
result bility<sup>2</sup> is  
s of the  
the varia  
R&R nce  
analy assoc  
sis iated  
for



to the nt  
 repea da  
 tabilit ta  
 y and for  
 is ea  
 $\sigma_{\text{reprodu}}$  ch  
 cibility<sup>2</sup> sta  
 the ge  
 varia pri  
 nce or  
 assoc to  
 iated est  
 to the ab  
 repro lis  
 ducib hi  
 ility ng  
 th  
 $\sigma_{\text{to}}$  e  
 tal2 an  
 $\frac{1}{4}$  al  
 $\sigma_{\text{s}}$  ysi  
 yst s  
 em<sup>2</sup> ph  
 $\rho$  as  
 $\sigma_{\text{r}}$  e.  
 epe **Fi**  
 ata **gu**  
 bili **re**  
 ty2 **5**  
 $\sigma_{\text{r}}$  dis  
 epr pl  
 odu ay  
 cibi s  
 lity th  
<sup>2</sup> e  
 N p-  
 or va  
 m lu  
 ali es  
 ty for  
 tes bo  
 tin ths  
 g ta  
 wa ge  
 s s,  
 co in  
 nd di  
 uc cat  
 te in  
 d g  
 on va  
 th lu  
 e es  
 cu gr  
 rre eat

---

er to  
th as  
an se  
0. ss  
05 pr  
. oc  
Th es  
is s  
su sta  
gg bil  
est ity  
s ov  
th er  
at ti  
th m  
e e.  
da M  
ta ini  
in ta  
ea b  
ch so  
sta ft  
ge wa  
fol re  
lo for  
ws pe  
a rfo  
no rm  
rm in  
al g  
dis th  
tri e  
bu ga  
tio ug  
n, e  
th R  
us &  
jus R  
tif stu  
yi dy  
ng ,  
th Ga  
e ge  
co R  
nst &  
ru R  
cti res  
on ult  
of s  
co for  
ntr th  
ol e  
ch po  
art w  
s de

---

r d  
sta th  
ge e  
an pe  
d rc  
th en  
e ta  
ta ge  
bl of  
et va  
sta ria  
ge tio  
s n  
ar ex  
e pl  
de ai  
pi ne  
cte d  
d by  
by th  
Fi e  
gu ov  
re er  
6. all  
Th va  
e ria  
pe bil  
rc ity  
en ,  
ta ar  
ge e  
of als  
va o  
ria an  
tio al  
n yz  
in ed  
th usi  
e ng  
da Mi  
ta nit  
att ab  
rib .  
ut Th  
ed e  
to res  
ov ult  
er s  
all ar  
va e  
ria pr  
bil es  
ity en  
, te  
an d

in  
Fi  
gu  
res  
7  
and  
8.

# **Gage R&R Study - ANOVA Method: Two-Way ANOVA Table With Interaction**

| Source        | DF | SS       | MS      | F       | P     |
|---------------|----|----------|---------|---------|-------|
| Batch no.     | 9  | 25338984 | 2815443 | 1735547 | 0.000 |
| Operator no.  | 1  | 32       | 32      | 20      | 0.002 |
| Repeatability | 9  | 15       | 2       | 1       | 0.394 |
| Total         | 20 | 29       | 1       |         |       |

Alpha to remove interaction term = 0.25

## **Two-Way ANOVA Table Without Interaction**

| Source        | DF | SS       | MS      | F       | P     |
|---------------|----|----------|---------|---------|-------|
| Batch no.     | 9  | 25338984 | 2815443 | 1872657 | 0.000 |
| Operator no.  | 1  | 32       | 32      | 22      | 0.000 |
| Repeatability | 29 | 44       | 2       |         |       |
| Total         | 39 | 25339060 |         |         |       |

## **Gage R&R**

| Source          | VarComp | %Contribution<br>(of VarComp) |
|-----------------|---------|-------------------------------|
| Total Gage R&R  | 3       | 0.00                          |
| Repeatability   | 2       | 0.00                          |
| Reproducibility | 2       | 0.00                          |
| Operator no.    | 2       | 0.00                          |
| Part-To-Part    | 703860  | 100.00                        |
| Total Variation | 703863  | 100.00                        |

| Source          | StdDev (SD) | Study Var<br>(6 * SD) | %Study Var<br>(%SV) |
|-----------------|-------------|-----------------------|---------------------|
| Total Gage R&R  | 1.746       | 10.48                 | 0.21                |
| Repeatability   | 1.226       | 7.36                  | 0.15                |
| Reproducibility | 1.243       | 7.46                  | 0.15                |
| Operator no.    | 1.243       | 7.46                  | 0.15                |
| Part-To-Part    | 838.964     | 5033.78               | 100.00              |
| Total Variation | 838.966     | 5033.79               | 100.00              |

Number of Distinct Categories = 677

For  
r the  
powd  
er  
stage,  
Minit  
ab  
analy  
sis  
sugg  
ests  
that

Fig  
ur  
e  
7.  
Th  
e  
Mi  
nit  
ab  
re  
sul  
t

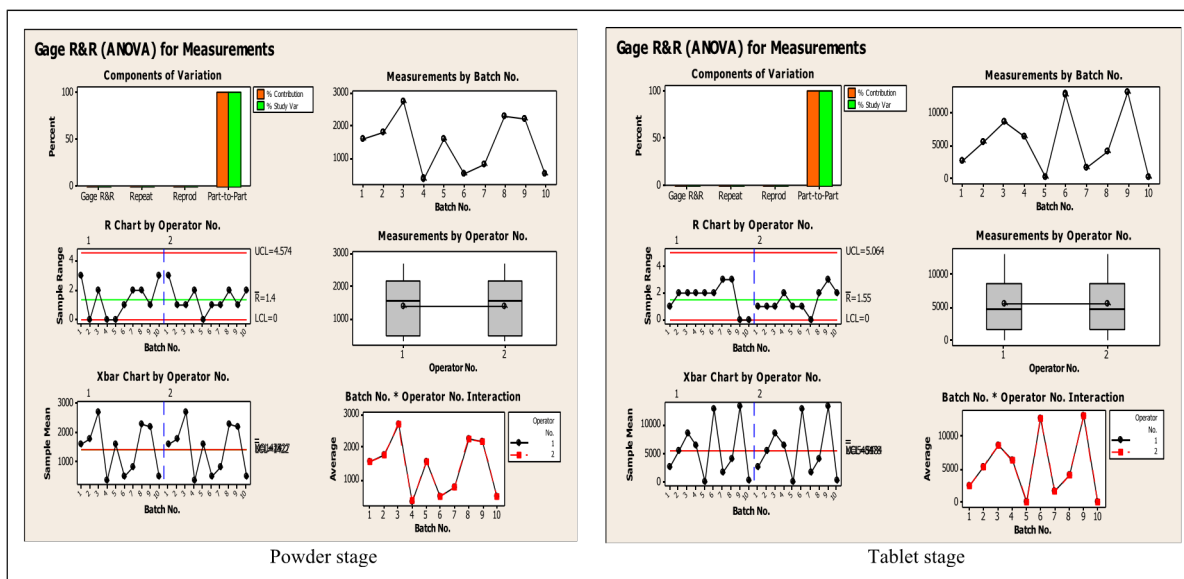


Figure 6. Gage R&R evaluation for the two stages.

the  
intera  
ction  
effect  
lacks

for  
ga  
ug  
e  
R

---

& va  
R ri  
of ab  
po ili  
w ty  
de ar  
r e  
st un  
ag w  
e. ar  
ra  
co nt  
nc ed  
er .  
ns U  
re po  
ga n  
rd ex  
in a  
g m  
ba in  
tc in  
h- g  
to th  
- e  
ba co  
tc nt  
h ri  
an bu  
d ti  
op on  
er pe  
at rc  
or en  
va ta  
ri ge  
ab ta  
ili bl  
ty e,  
in it'  
re s  
du ev  
ci id  
ng en  
m t  
ea th  
su at  
re th  
m e  
en to  
t tal  
sy ga  
st ge  
e R  
m &

---

R m  
co in  
nt g  
ri th  
bu e  
ti ad  
on eq  
pe ua  
rc cy  
en of  
ta th  
ge e  
re sy  
m st  
ai e  
ns m.  
be A  
lo dd  
w iti  
1 on  
% all  
, y,  
wi th  
th e  
bo va  
th ri  
re an  
pe ce  
at pe  
ab rc  
ili en  
ty ta  
an ge  
d ta  
re bl  
pr e  
od re  
uc ve  
ib al  
ili s  
ty th  
co at  
nt th  
ri e  
bu to  
ti tal  
on ga  
s ge  
at R  
0 &  
% R  
, va  
co ri  
nfi an  
r ce

pe  
rc  
en  
ta  
ge  
re  
pr  
es  
en  
ts  
on  
ly  
0.  
21  
%  
of  
th  
e  
ov  
er  
all  
va  
ri  
ab  
ili  
ty,  
co  
m  
fo  
rt

ab  
ly  
be  
lo  
w  
th  
e  
ac  
ce  
pt  
ab  
le  
th  
re  
sh  
ol  
d  
of  
10  
%  
.  
B  
ot  
h  
re  
pe  
at  
ab  
ili  
ty

#### Gage R&R Study - ANOVA Method: Two-Way ANOVA Table With Interaction

| Source                   | DF | SS        | MS       | F        | P     |
|--------------------------|----|-----------|----------|----------|-------|
| Batch no.                | 9  | 822987753 | 91443084 | 94325244 | 0.000 |
| Operator no.             | 1  | 11        | 11       | 11       | 0.008 |
| Batch no. * Operator no. | 9  | 9         | 1        | 1        | 0.785 |
| Repeatability            | 20 | 33        | 2        |          |       |
| Total                    | 39 | 822987805 |          |          |       |

Alpha to remove interaction term = 0.25

#### Two-Way ANOVA Table Without Interaction

| Source        | DF | SS        | MS       | F        | P     |
|---------------|----|-----------|----------|----------|-------|
| Batch no.     | 9  | 822987753 | 91443084 | 64326244 | 0.000 |
| Operator no.  | 1  | 11        | 11       | 8        | 0.009 |
| Repeatability | 29 | 41        | 1        |          |       |
| Total         | 39 | 822987805 |          |          |       |

#### Gage R&R

| Source          | VarComp  | %Contribution<br>(of VarComp) |
|-----------------|----------|-------------------------------|
| Total Gage R&R  | 2        | 0.00                          |
| Repeatability   | 1        | 0.00                          |
| Reproducibility | 0        | 0.00                          |
| Operator no.    | 0        | 0.00                          |
| Part-To-Part    | 22860771 | 100.00                        |
| Total Variation | 22860772 | 100.00                        |

| Source          | StdDev (SD) | Study Var<br>(6 * SD) | %Study Var<br>(%SV) |
|-----------------|-------------|-----------------------|---------------------|
| Total Gage R&R  | 1.38        | 8.3                   | 0.03                |
| Repeatability   | 1.19        | 7.2                   | 0.02                |
| Reproducibility | 0.69        | 4.2                   | 0.01                |
| Operator no.    | 0.69        | 4.2                   | 0.01                |
| Part-To-Part    | 4781.29     | 28687.8               | 100.00              |
| Total Variation | 4781.29     | 28687.8               | 100.00              |

Number of Distinct Categories = 4888

Figure 8.  
The Minitab  
result for

gauge R & R  
of tablet  
stage.

Figure 9. The  
Laney P<sup>0</sup> chart  
for the  
defective units  
performed with  
unequal sample  
size.

and as  
repro batch  
ducib numb  
ility er  
value and  
s, opera  
each tor  
at identi  
0.15 ty,  
%, lacke  
furthe d  
r statist  
endor ical  
se the signi  
precis ficanc  
ion of e (p =  
the .785),  
meas sugge  
urem sting  
ent their  
devic negli  
e. gible

In influe  
the nce  
tablet on  
stage meas  
of our urem  
study, ent  
the syste  
ANO m  
VA varia  
table bility.  
reveal More  
ed over,  
that the  
the table  
intera disclo  
ction sed a  
effect repeat  
, abilit  
invol y  
ving degre  
factor e of  
s freed  
such om of



---

20, syste  
indica m.  
ting Furth  
the ermor  
numb e,  
er of exami  
repeat natio  
meas n of  
urem the  
ents perce  
perfor ntage  
med. study  
Analy varia  
sis of nce  
the table  
perce highli  
ntage ghted  
contri that  
butio the  
n total  
table gage  
indica R&R  
ted varia  
that nce  
both consti  
total tuted  
gage only  
R&R 0.03  
contri % of  
butio the  
n and overa  
indivi ll  
dual varia  
repeat nce,  
abilit comf  
y and ortabl  
repro y  
ducib below  
ility the  
contri accep  
butio table  
ns thresh  
remai old of  
ned 10%.  
below Both  
1%, repeat  
valida abilit  
ting y and  
the repro  
adequ ducib  
acy ility  
of the value  
meas s fell  
urem withi  
ent n

acceptance

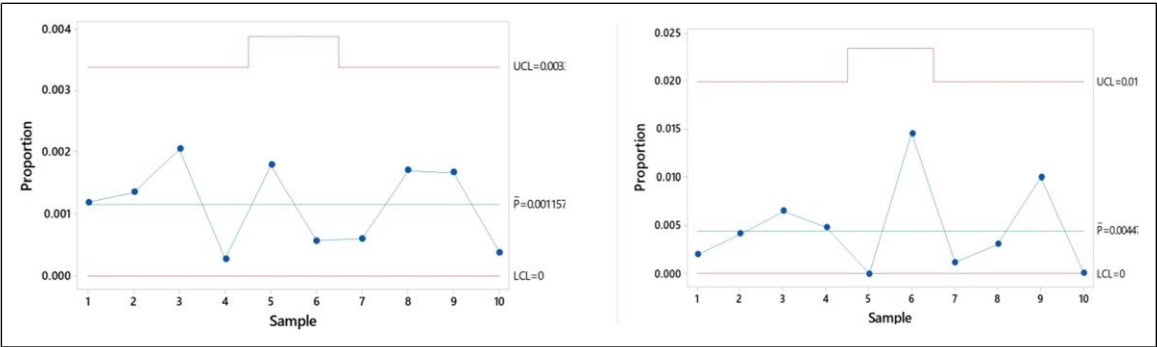


table adjust  
range ments  
s. for  
Co both  
nsider over-  
ing disper  
the sion  
consi and  
derab under  
le -  
varia disper  
bility sion  
in withi  
sampl n the  
e datas  
sizes et.  
withi Over-  
n the disper  
curre sion  
nt can  
data result  
set, in an  
the increa  
consi sed  
derati numb  
on of er of  
a p points  
prime outsid  
chart, e the  
rather contr  
than a ol  
traditi limits  
onal in a  
p traditi  
chart, onal  
is p  
warra chart,  
nted. while  
This under  
variati -  
ion disper  
neces sion  
sitate may

lead to too few points outside

identi-  
fication of  
unstable  
points

| Stage  | P-chart |         |     | Decision   | Capability estimations |                |           |         |      |      | Decision |
|--------|---------|---------|-----|------------|------------------------|----------------|-----------|---------|------|------|----------|
|        | UCL     | CL      | LCL |            | DPMO                   | Difference (%) | Yield (%) | Z-value | SQL  | Cp   |          |
| Powder | 0.00338 | 0.00115 | 0   | In-control | 1157.13                | 0.12           | 99.88     | 3.046   | 4.58 | 1.53 | Capable  |
| Tablet | 0.01988 | 0.00447 | 0   | In-control | 4465.57                | 0.45           | 99.55     | 2.614   | 4.17 | 1.39 | Capable  |

these limits. The Lane

y P<sup>0</sup>

chart

effect

ively

addre

sses

these

challe

nges

by

acco

mmo

datin

g

such

condi

tions.

Comp

ared

to the

stand

ard p

chart,

the p

prime

chart

offers

wider

and

varyi

ng

contr

ol

limits

,

facilit

ating

the

withi  
n the  
proce  
ss.

Tabl  
e 3.

Sum  
mar

y of  
stati

stica

l  
resul

ts of  
mea

sure  
men

t  
phas

e.  
F

ig  
ur

e 9  
de

pic  
ts

the  
im

ple  
me

nta  
tio

n  
of

the  
La

ne  
y

P<sup>0</sup>

---

ch pr  
art oc  
for ess  
de sta  
fec bil  
tiv ity  
e an  
un d  
its ide  
at nti  
the fyi  
po ng  
wd po  
er ten  
an tia  
d l  
tab ou  
let tli  
sta ers  
ge or  
s, un  
res sta  
pe ble  
cti po  
vel int  
y, s  
pr ty  
ov pic  
idi al  
ng fro  
a m  
co co  
m m  
pr m  
eh on  
en ca  
siv us  
e es.  
vis P  
ual ro  
re ce  
pr ss  
es ca  
ent pa  
ati bil  
on ity  
tha an  
t aly  
aid sis  
s is  
in a  
ass pi  
ess vo  
in tal  
g as

---

pe ort  
 ct io  
 of n  
 Si of  
 x de  
 Si fec  
 g tiv  
 ma e  
 me ite  
 th ms  
 od be  
 ol co  
 og me  
 y. s  
 Wi fea  
 th sib  
 the le.  
 pr Ke  
 oc y  
 ess me  
 co tri  
 nfi cs  
 rm su  
 ed ch  
 as as  
 sta the  
 ble Z-  
 , val  
 in ue,  
 dic re  
 ate pr  
 d es  
 by ent  
 the in  
 p<sup>o</sup> g  
 ch sta  
 art nd  
 's ar  
 sta d  
 tist de  
 ica via  
 l tio  
 co ns  
 ntr fro  
 ol, m  
 est the  
 im me  
 ati an,  
 on an  
 of d  
 the the  
 tru de  
 e fec  
 pr t  
 op rat

---

|      |     |
|------|-----|
| e    | ce  |
| pe   | d   |
| r    | err |
| mi   | ors |
| lli  | in  |
| on   | pr  |
| (P   | od  |
| P    | uct |
| M)   | io  |
| or   | n   |
| de   | or  |
| fec  | ser |
| t    | vic |
| pe   | e   |
| r    | pr  |
| mi   | oc  |
| lli  | ess |
| on   | es. |
| op   | Ca  |
| po   | lcu |
| rtu  | lat |
| nit  | io  |
| ies  | ns  |
| (D   | of  |
| P    | Cp  |
| M    | at  |
| O)   | sig |
| aid  | ma  |
| in   | lev |
| ass  | els |
| ess  | of  |
| in   | 4.5 |
| g    | 8   |
| pr   | for |
| oc   | po  |
| ess  | wd  |
| ca   | er  |
| pa   | sta |
| bil  | ge  |
| ity. | an  |
| Lo   | d   |
| we   | 4.1 |
| r    | 7   |
| sig  | for |
| ma   | tab |
| lev  | let |
| els  | sta |
| co   | ge  |
| rre  | co  |
| lat  | nfi |
| e    | rm  |
| wi   | pr  |
| th   | oc  |
| re   | ess |
| du   | ca  |

---

pa in  
 bil Ta  
 ity, ble  
 wi 3.  
 th B  
 val y  
 ue ap  
 s pl  
 ex yi  
 ce ng  
 edi Sh  
 ng ew  
 1 ha  
 in rt  
 dic rul  
 ati es  
 ng to  
 ca the  
 pa p<sup>0</sup>  
 bil co  
 ity. ntr  
 St ol  
 ati ch  
 sti art  
 cal m  
 dat on  
 a ito  
 fro rin  
 m g  
 the de  
 me fec  
 as tiv  
 ur e  
 em un  
 ent its,  
 ph it  
 as co  
 e nfi  
 for rm  
 po s  
 wd tha  
 er t  
 an ea  
 d ch  
 tab sta  
 let ge  
 sta of  
 ge the  
 s pr  
 are oc  
 su ess  
 m is  
 ma sta  
 riz ble  
 ed an

---

d un  
un de  
de rly  
r in  
co g  
ntr rea  
ol. so  
Th ns  
is be  
in hi  
dic nd  
ate no  
s nc  
tha on  
t for  
the mi  
ob tie  
ser s  
ve an  
d d  
va th  
ria or  
tio ou  
n gh  
in ly  
de ex  
fec am  
tiv ine  
e all  
un po  
its ten  
ali tia  
gn l  
s ca  
wi us  
th es.  
wh Th  
at is  
is pr  
Ro oc  
ot ess  
Ca fac  
us ilit  
e ate  
An s  
aly the  
sis dis  
ai co  
ms ve  
to ry  
un of  
co sol  
ve uti  
r on  
the s



---

an  
d  
the  
im  
ple  
me  
nta  
tio  
n  
of  
co  
rre  
cti  
ve  
act  
io  
ns.  
[Fi](#)  
[gu](#)  
[re](#)  
[10](#)  
pr  
es  
ent  
s a  
fis  
hb  
on  
e  
dia  
gr  
am  
tha  
t  
ill  
ust  
rat  
es  
the  
ass  
oci  
ate  
d  
ro  
ot  
ca  
us  
es  
in  
the  
tab  
let  
co  
m  
pr

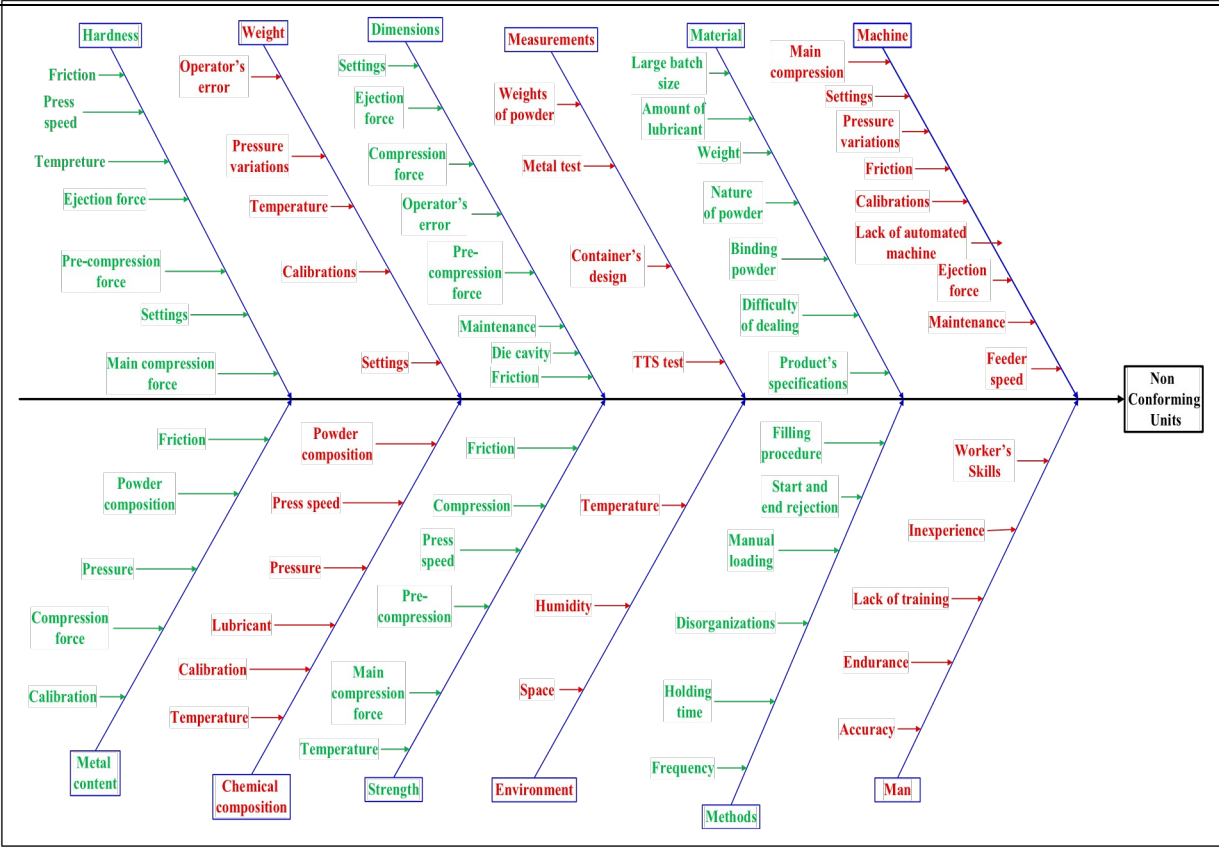


Figure 10. Fishbone diagram associated with nonconformities in the tablet compression process.

Bo Why  
th there  
stage are a  
s nonc  
under onfor  
went ming  
Five powd  
Whys er  
analy units  
sis. ? 2.  
The Why  
Five there  
Whys is a  
inves large  
tigati wasta  
on of ge of  
nonc powd  
onfor er  
mitie units  
s at durin  
the g  
powd loadi  
er ng  
stage into  
unco the  
vered comp  
: 1. ressi

---

|      |      |
|------|------|
| n    | r    |
| m    | w    |
| ac   | or   |
| hi   | ke   |
| ne   | rs   |
| ?    | ?    |
| 3.   | 5.   |
| W    | W    |
| hy   | hy   |
| is   | th   |
| th   | er   |
| e    | e    |
| fill | ar   |
| in   | e    |
| g    | es   |
| pr   | se   |
| oc   | nti  |
| ed   | al   |
| ur   | au   |
| e    | to   |
| m    | m    |
| an   | at   |
| ua   | ed   |
| l    | fill |
| fo   | in   |
| r    | g    |
| w    | to   |
| or   | ol   |
| ke   | s,   |
| rs   | lik  |
| ?    | e    |
| 4.   | th   |
| W    | e    |
| hy   | va   |
| is   | cu   |
| th   | u    |
| e    | m    |
| fill | lo   |
| in   | ad   |
| g    | er,  |
| m    | ov   |
| et   | erl  |
| ho   | oo   |
| d    | ke   |
| ex   | d.   |
| ce   | Fi   |
| ssi  | ve   |
| ve   | W    |
| ly   | hy   |
| ch   | s    |
| all  | in   |
| en   | ve   |
| gi   | sti  |
| ng   | ga   |
| fo   | tio  |

|     |      |
|-----|------|
| n   | cu   |
| int | r    |
| o   | at   |
| ta  | ba   |
| bl  | tc   |
| et  | h    |
| st  | st   |
| ag  | art  |
| e   | an   |
| no  | d    |
| nc  | en   |
| on  | d?   |
| fo  | 3.   |
| r   | W    |
| mi  | hy   |
| tie | m    |
| s   | ac   |
| re  | hi   |
| ve  | ne   |
| al  | se   |
| s:  | titi |
| 1.  | ng   |
| W   | s    |
| hy  | ar   |
| th  | e    |
| er  | fr   |
| e   | eq   |
| ar  | ue   |
| e   | ntl  |
| no  | y    |
| n-  | ch   |
| co  | an   |
| nf  | ge   |
| or  | d    |
| mi  | pe   |
| ng  | r    |
| ta  | ba   |
| bl  | tc   |
| et  | h?   |
| un  | 4.   |
| its | W    |
| ?   | hy   |
| 2.  | th   |
| W   | e    |
| hy  | m    |
| do  | an   |
| pr  | uf   |
| es  | ac   |
| su  | tu   |
| re  | ri   |
| ch  | ng   |
| an  | or   |
| ge  | de   |
| s   | r    |
| oc  | is   |

---

|     |      |
|-----|------|
| di  | et   |
| so  | co   |
| rg  | m    |
| an  | pr   |
| iz  | es   |
| ed  | si   |
| ?   | on   |
| 5.  | pr   |
| W   | oc   |
| hy  | es   |
| th  | s    |
| er  | ai   |
| e   | m    |
| is  | ed   |
| no  | to   |
| or  | ad   |
| ga  | dr   |
| ni  | es   |
| ze  | s    |
| d   | id   |
| sc  | en   |
| he  | tifi |
| du  | ed   |
| lin | iss  |
| g   | ue   |
| of  | s    |
| or  | an   |
| de  | d    |
| rs. | en   |
|     | ha   |
| I   | nc   |
| m   | e    |
| p   | ov   |
| r   | er   |
| o   | all  |
| v   | ef   |
| e   | fic  |
| T   | ie   |
| he  | nc   |
| I   | y    |
| m   | an   |
| pr  | d    |
| ov  | pr   |
| e   | od   |
| ph  | uc   |
| as  | t    |
| e   | qu   |
| of  | ali  |
| th  | ty.  |
| e   | C    |
| ta  | ha   |
| bl  | lle  |
|     | ng   |
|     | es   |
|     | id   |

---

en s,  
tifi in  
ed ad  
in eq  
cl ua  
ud te  
ed pr  
fr ev  
eq en  
ue tiv  
nt e  
se an  
tti d  
ng co  
ad rr  
ju ec  
st tiv  
m e  
en m  
ts ai  
du nt  
e en  
to an  
ta ce  
bl m  
et ea  
rej su  
ec re  
tio s,  
n an  
de d  
fa in  
ult su  
s, ffi  
he ci  
ig en  
ht t  
en w  
ed or  
fri ke  
cti r  
on tra  
le ini  
ad ng  
in on  
g m  
to ac  
m hi  
et ne  
al se  
te tti  
st ng  
fai s.  
lu T  
re he

---

|     |      |
|-----|------|
| se  | su   |
| fa  | ch   |
| ct  | as   |
| or  | ac   |
| s   | tiv  |
| si  | ati  |
| gn  | ng   |
| ifi | au   |
| ca  | to   |
| ntl | m    |
| y   | at   |
| co  | ed   |
| nt  | fill |
| ri  | in   |
| bu  | g    |
| te  | pr   |
| d   | oc   |
| to  | ed   |
| de  | ur   |
| fe  | es,  |
| ct  | op   |
| s   | ti   |
| in  | mi   |
| th  | zi   |
| e   | ng   |
| pr  | m    |
| od  | an   |
| uc  | uf   |
| tio | ac   |
| n   | tu   |
| pr  | ri   |
| oc  | ng   |
| es  | or   |
| s.  | de   |
| B   | r    |
| y   | sc   |
| im  | he   |
| pl  | du   |
| e   | lin  |
| m   | g,   |
| en  | str  |
| tin | ict  |
| g   | ly   |
| er  | ad   |
| ro  | he   |
| r-  | ri   |
| pr  | ng   |
| oo  | to   |
| fin | S    |
| g   | O    |
| str | Ps   |
| at  | ,    |
| eg  | en   |
| ie  | su   |
| s   | ri   |

---

ng  
pr  
op  
er  
lu  
br  
ic  
ati  
on  
,  
an  
d  
pr  
ov  
idi  
ng  
co  
m  
pr  
eh  
en  
si  
ve  
w  
or  
ke  
r  
tra  
ini  
ng  
,  
su  
bs  
ta  
nti  
al  
im  
pr  
ov  
e  
m  
en  
ts  
w  
er  
e  
ac  
hi  
ev  
ed  
.   
T  
he  
se  
so  
lut  
io  
ns  
no  
t  
on  
ly  
ad  
dr  
es  
se  
d  
id  
en  
tifi  
ed  
iss  
ue  
s  
bu  
t  
al  
so  
le  
d  
to  
re  
m  
ar  
ka  
bl  
e  
en  
ha  
nc  
e  
m  
en  
ts  
in  
pr  
oc  
es  
s  
ef  
fic  
ie  
nc  
y  
an  
d  
pr  
od  
uc  
t



---

qu capab  
 ali ility,  
 ty. sigma  
 Fo level  
 llowi associ  
 ng ated  
 these with  
 strate impro  
 gies veme  
 and nts,  
 analy and  
 sis of others  
 newly , were  
 gathe then  
 red reeval  
 sampl uated.  
 e A  
 data, befor  
 revisi e-  
 ons after  
 were comp  
 made arison  
 to was  
 contr condu  
 ol cted,  
 charts as  
 (see illustr  
 Figur ated  
 e 11) in  
 for Table  
 nonco 4.  
 nform These  
 ities error-  
 in the proofi  
 powd ng  
 er strate  
 stage. gies  
 Key effect  
 metri ively  
 cs, reduc  
 inclu ed  
 ding errors  
 Z- and  
 value, defect  
 Defec s in  
 ts Per the  
 Milli produ  
 on ction  
 Oppo proce  
 rtuniti ss,  
 es result  
 (DP ing in  
 MO), impro  
 proce ved  
 ss proce

ss  
effici  
ency  
and  
produ  
ct  
qualit  
y.  
  
Co  
ntr  
ol  
  
The  
Contr  
ol

soluti  
ons.  
This  
phase  
entail  
ed  
the  
rigor  
ous  
imple  
ment  
ation  
of  
contr  
ols,  
inclu  
ding

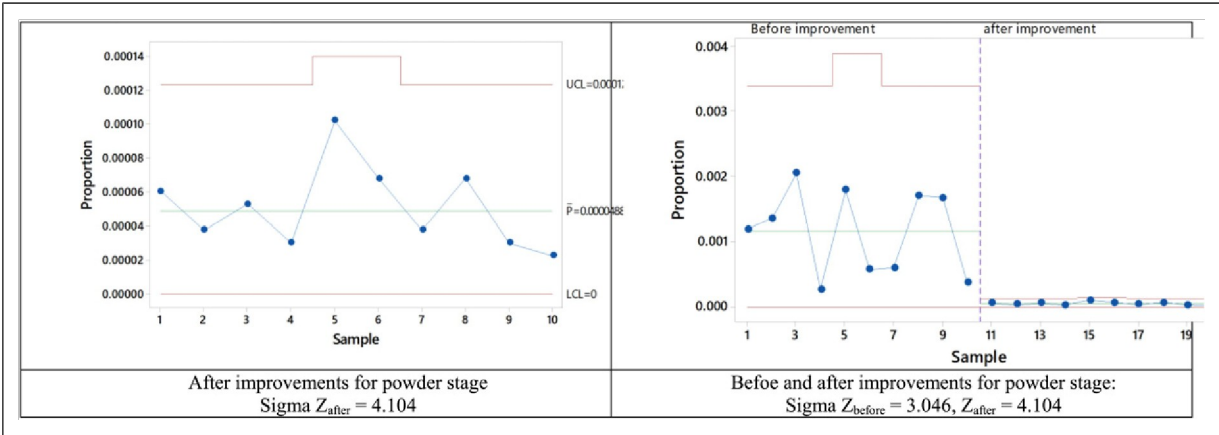


Figure 11. The Laney P<sup>0</sup> chart for number of nonconformities before and after improvements.

phase  
of the  
DMA  
IC  
proce  
ss has  
been  
metic  
ulous  
ly  
execu  
ted to  
ensur  
e the  
lastin  
g  
effect  
iveness  
of  
imple  
ment  
ed

the  
updat  
ing of  
Stand  
ard  
Oper  
ating  
Proce  
dures  
(SOP  
s)  
and  
target  
ed  
traini  
ng  
initiat  
ives  
aimed  
at  
foster  
ing

---

consi manc  
 stent e.  
 adher These  
 ence syste  
 to ms  
 optim enabl  
 ized ed  
 proce prom  
 sses. pt  
 Co identi  
 ntinu ficati  
 ous on of  
 Monit deviat  
 oring ions  
 for from  
 Perfo establ  
 rman ished  
 ce stand  
 Evalu ards,  
 ation: allow  
 Sophi ing  
 sticat for  
 ed proac  
 monit tive  
 oring interv  
 syste entio  
 ms, n to  
 such maint  
 as ain  
 contr qualit  
 ol y  
 charts levels  
 and .  
 statist Pr  
 ical eventi  
 proce ng  
 ss Recur  
 contr rence  
 ol of  
 (SPC) Defec  
 techni ts:  
 ques, Proac  
 were tive  
 adeptl meas  
 y ures  
 deplo were  
 yed taken  
 to to  
 conti preve  
 nuous nt the  
 ly recurr  
 assess ence  
 proce of  
 ss previ  
 perfor ously

identi mitm  
 fied ent to  
 defect conti  
 s or nuous  
 issues impro  
 . This veme  
 invol nt  
 ved a was  
 focus demo  
 ed nstrat  
 effort ed  
 on throu  
 addre gh  
 ssing the  
 poten establ  
 tial ishme  
 sourc nt of  
 es of robus  
 variat t  
 ion or feedb  
 non- ack  
 confo mech  
 rman anism  
 ce s.  
 befor These  
 e they mech  
 could anism  
 impac s  
 t facilit  
 produ ated  
 ct ongoi  
 qualit ng  
 y, refine  
 thus ment  
 ensuri and  
 ng a optim  
 sustai izatio  
 ned n of  
 impro proce  
 veme sses,  
 nt reflec  
 trajec ting  
 tory. an  
 Co organ  
 mmit izatio  
 ment nal  
 to ethos  
 Ongo geare  
 ing d  
 Impro towar  
 veme ds  
 nt: A perpe  
 steadf tual  
 ast enhan  
 com

---

ceme learni  
 nt. ng,  
 Tr contri  
 anspa butin  
 rent g  
 Docu to  
 menta a  
 tion  
 and cultur  
 Repor e  
 ting: of  
 Comp data-  
 rehen drive  
 sive n  
 docu  
 menta qualit  
 tion y  
 and mana  
 report geme  
 ing nt.  
 proce T  
 dures hr  
 were ou  
 instit gh  
 uted the  
 to dil  
 ensur ige  
 e nt  
 transp ex  
 arenc ec  
 y and uti  
 accou on  
 ntabil of  
 ity the  
 throu Co  
 ghout ntr  
 the ol  
 Contr ph  
 ol as  
 phase e,  
 . This the  
 frame res  
 work ear  
 facilit ch  
 ated ca  
 infor se  
 med ex  
 decisi em  
 on- pli  
 maki fie  
 ng s a  
 and pr  
 organ oa  
 izatio cti  
 nal ve

ap  
pr  
oa  
ch  
to  
qu  
ali  
ty  
ma  
na  
ge  
me  
nt,  
un  
de  
rpi  
nn  
ed  
by  
a  
co  
m  
mi  
tm  
ent  
to  
co  
nti  
nu  
ou  
s  
im  
pr  
ov  
em  
ent  
pri  
nci  
ple  
s.

Ov  
erc  
o  
mi  
ng  
ch  
all  
en  
ge  
s  
an  
d  
en  
sur  
ing  
sus  
tai  
na  
ble  
im  
pr  
ov  
em  
en  
t  
Ap  
pl  
yi  
ng  
Le  
an  
Si  
x  
Si  
g  
ma  
D  
M  
AI  
C  
in  
ph  
ar  
ma  
ce  
uti

---

cal s,  
ma an  
nu d  
fac va  
tur ria  
in bil  
g ity  
led in  
to ma  
sig chi  
ni ne  
fic set  
ant tin  
im gs,  
pr alo  
ov ng  
em wi  
ent th  
s. the  
Ho sol  
we uti  
ver on  
, s  
ma us  
in ed  
ch to  
all ov  
en erc  
ge o  
s me  
fac the  
ed m,  
du thi  
rin s  
g inf  
ea or  
ch ma  
su tio  
ch n  
as is  
res su  
ist m  
an ma  
ce riz  
to ed  
ch in  
an [Ta](#)  
ge, [ble](#)  
dat [5.](#)  
a  
inc  
on  
sis  
ten  
cie

---

Re  
sul  
ts  
an  
d  
dis  
cu  
ssi  
on  
Th  
is  
pa  
pe  
r  
ex  
pl  
or  
es  
the  
us  
e  
of  
the  
D  
M  
AI  
C  
ap  
pr  
oa  
ch  
wi  
thi  
n  
ph  
ar  
ma  
ce  
uti  
cal  
pr  
od  
uct  
io  
n,  
ex  
am  
ini  
ng  
ho  
w  
tar

get  
ed  
im  
pr  
ov  
em  
ent  
s  
acr  
os  
s  
ea  
ch  
ph  
as  
e  
in  
flu  
en  
ce  
d  
ov  
era  
ll  
pr  
oc  
ess  
pe  
rfo  
rm  
an  
ce.  
E  
m  
ph  
asi  
s  
wa  
s  
pla  
ce  
d  
on  
me  
as  
ur  
abl  
e  
ou  
tco  
me  
s,  
co  
m  
pa



ris g  
on ma  
s pri  
wi nci  
th ple  
pri s  
or thr  
res ou  
ear gh  
ch, the  
an D  
d M  
the AI  
pr C  
act str  
ica uct  
l ur  
val e  
ue br  
the ou  
se gh  
ch t  
an ab  
ge ou  
s t  
ma ma  
y rk  
off ed  
er en  
to ha  
the nc  
in em  
du ent  
str s  
y. in  
Ap co  
pl nsi  
yi ste  
ng nc  
Le y,  
an qu  
Si ali  
x ty,  
Si

Table 4. The statistical data obtained for powder stage before and after improvement.

| Stage | P <sup>1</sup> -chart |     |  | Decision | Capability estimations |        |         |                          |     |    |                      |
|-------|-----------------------|-----|--|----------|------------------------|--------|---------|--------------------------|-----|----|----------------------|
|       | UCL                   |     |  |          | DPMO                   | Def. % | Yield % | Z-value<br>(sigma level) | SQL | Cp | Capability statement |
|       | CL                    | LCL |  |          |                        |        |         |                          |     |    |                      |

|        |          |           |   |            |         |       |        |       |      |      |                     |
|--------|----------|-----------|---|------------|---------|-------|--------|-------|------|------|---------------------|
| Before | 0.00338  | 0.00115   | 0 | In-control | 1157.13 | 0.12  | 99.88  | 3.046 | 4.58 | 1.53 | Capable             |
| After  | 0.000012 | 0.0000488 | 0 | In-control | 48.79   | 0.005 | 99.995 | 4.104 | 5.4  | 1.8  | Improved capability |

Table 5. The challenges faced during each DMAIC phase and how they were overcome.

| DMAIC phase | Key challenges                                                                                      | Overcoming method                                                                                                |
|-------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Define      | Employees appeared reluctant to adopt six sigma, fearing increased workload or job security threats | - Facilitated focus group discussions, to engage workforces in the process<br>- Offer six sigma training courses |
|             | Identifying the critical process-causing non-conformance                                            | - Using SIPOC charts to clarify stakeholders                                                                     |
| Measure     | Variations in manual data recording led to errors                                                   | - Ensuring measurement system reliability by conducting repeatability and reproducibility                        |
| Analyze     | Multiple factors influenced non-conformance, making it difficult to find the primary causes         | - Using fishbone charts and five whys to break down the issues scientifically                                    |
| Improve     | Operators frequently changed settings, leading to nonconformance                                    | - Automated the filling process to reduce errors                                                                 |
| Control     | Employees might revert to old practices                                                             | - Continuous monitoring via SPC                                                                                  |
|             | Difficulty in ensuring long-term stability of the improvements                                      | - Rewarded workforces for following to process improvements targets                                              |

an  
d  
op  
er  
ati  
on  
al  
ef  
fic  
ie  
nc  
y.  
In  
th  
e  
in  
iti  
al  
D  
efi  
ne  
ph  
as  
e,  
ke  
y  
so  
ur  
ce  
s  
of  
no  
n-  
co  
nf  
or  
m  
an  
ce  
w  
er  
e  
id  
en  
ti  
fie  
d,  
ai  
de  
d  
by  
SI

---

|     |     |
|-----|-----|
| P   | lia |
| O   | bi  |
| C   | lit |
| di  | y,  |
| ag  | us  |
| ra  | in  |
| m   | g   |
| s   | R   |
| th  | ep  |
| at  | ea  |
| cl  | ta  |
| ar  | bi  |
| ifi | lit |
| ed  | y   |
| pr  | an  |
| oc  | d   |
| es  | R   |
| s   | ep  |
| bo  | ro  |
| un  | du  |
| da  | ci  |
| ri  | bi  |
| es  | lit |
| an  | y   |
| d   | (R  |
| st  | &   |
| ak  | R)  |
| eh  | an  |
| ol  | al  |
| de  | ys  |
| r   | is  |
| re  | to  |
| lat | ad  |
| io  | dr  |
| ns  | es  |
| hi  | s   |
| ps  | in  |
| .   | co  |
| T   | ns  |
| he  | ist |
| M   | en  |
| ea  | ci  |
| su  | es  |
| re  | an  |
| ph  | d   |
| as  | va  |
| e   | li  |
| fo  | da  |
| cu  | te  |
| se  | th  |
| d   | e   |
| on  | m   |
| da  | ea  |
| ta  | su  |
| re  | re  |

---

|     |     |
|-----|-----|
| m   | ze  |
| en  | st  |
| t   | ag  |
| sy  | e,  |
| st  | m   |
| e   | ul  |
| m.  | ti  |
| T   | pl  |
| he  | e   |
| se  | ca  |
| st  | us  |
| ep  | es  |
| s   | of  |
| lai | de  |
| d   | fe  |
| a   | ct  |
| so  | s   |
| li  | w   |
| d   | er  |
| fo  | e   |
| un  | in  |
| da  | ve  |
| ti  | sti |
| on  | ga  |
| fo  | te  |
| r   | d   |
| da  | th  |
| ta  | ro  |
| -   | ug  |
| dr  | h   |
| iv  | str |
| en  | uc  |
| de  | tu  |
| ci  | re  |
| si  | d   |
| on  | to  |
| s   | ol  |
| m   | s   |
| ov  | li  |
| in  | ke  |
| g   | th  |
| fo  | e   |
| r   | Fi  |
| w   | sh  |
| ar  | bo  |
| d.  | ne  |
| D   | di  |
| ur  | ag  |
| in  | ra  |
| g   | m   |
| th  | an  |
| e   | d   |
| A   | th  |
| na  | e   |
| ly  | Fi  |

---

|     |     |
|-----|-----|
| ve  | bs  |
| W   | eq  |
| hy  | ue  |
| s   | nt  |
| te  | ph  |
| ch  | as  |
| ni  | e   |
| qu  | in  |
| e.  | cl  |
| T   | ud  |
| hi  | ed  |
| s   | au  |
| he  | to  |
| lp  | m   |
| ed  | ati |
| un  | on  |
| co  | in  |
| ve  | th  |
| r   | e   |
| th  | fil |
| e   | li  |
| un  | ng  |
| de  | pr  |
| rl  | oc  |
| yi  | es  |
| ng  | s,  |
| co  | w   |
| nt  | hi  |
| ri  | ch  |
| bu  | pl  |
| to  | ay  |
| rs  | ed  |
| to  | a   |
| va  | cr  |
| ri  | iti |
| ab  | ca  |
| ili | l   |
| ty. | ro  |
| I   | le  |
| m   | in  |
| pr  | m   |
| ov  | in  |
| e   | i   |
| m   | m   |
| en  | izi |
| ts  | ng  |
| in  | hu  |
| tr  | m   |
| od  | an  |
| uc  | er  |
| ed  | ro  |
| in  | r   |
| th  | an  |
| e   | d   |
| su  | st  |

---

|     |     |
|-----|-----|
| an  | tr  |
| da  | ol  |
| rd  | (S  |
| izi | P   |
| ng  | C)  |
| ou  | ,   |
| tp  | en  |
| ut. | su  |
| Fi  | ri  |
| na  | ng  |
| lly | th  |
| ,   | at  |
| th  | th  |
| e   | e   |
| C   | ga  |
| on  | in  |
| tr  | s   |
| ol  | m   |
| ph  | ad  |
| as  | e   |
| e   | w   |
| i   | er  |
| m   | e   |
| pl  | m   |
| e   | ai  |
| m   | nt  |
| en  | ai  |
| te  | ne  |
| d   | d   |
| re  | an  |
| al  | d   |
| -   | de  |
| ti  | vi  |
| m   | ati |
| e   | on  |
| m   | s   |
| on  | pr  |
| it  | o   |
| or  | m   |
| in  | pt  |
| g   | ly  |
| us  | ad  |
| in  | dr  |
| g   | es  |
| St  | se  |
| ati | d.  |
| sti | T   |
| ca  | o   |
| l   | va  |
| Pr  | li  |
| oc  | da  |
| es  | te  |
| s   | th  |
| C   | e   |
| on  | ef  |

---

|     |     |
|-----|-----|
| fe  | ag  |
| cti | e   |
| ve  | sa  |
| ne  | w   |
| ss  | a   |
| of  | dr  |
| th  | op  |
| es  | in  |
| e   | D   |
| in  | P   |
| te  | M   |
| rv  | O   |
| en  | fr  |
| ti  | o   |
| on  | m   |
| s,  | 11  |
| pe  | 57  |
| rf  | .1  |
| or  | 3   |
| m   | to  |
| an  | 48  |
| ce  | .7  |
| in  | 9,  |
| di  | w   |
| ca  | hi  |
| to  | le  |
| rs  | th  |
| w   | e   |
| er  | si  |
| e   | g   |
| tr  | m   |
| ac  | a   |
| ke  | le  |
| d   | ve  |
| th  | l   |
| ro  | ro  |
| ug  | se  |
| ho  | fr  |
| ut  | o   |
| th  | m   |
| e   | 3.  |
| pa  | 04  |
| pe  | 6   |
| r.  | to  |
| N   | 5.  |
| ot  | 4.  |
| ab  | T   |
| ly, | he  |
| th  | se  |
| e   | re  |
| po  | su  |
| w   | lts |
| de  | sp  |
| r   | ea  |
| st  | k   |

to re  
th st  
e an  
de d  
pt ou  
h t.  
of B  
i yr  
m ne  
pr et  
ov al.  
e (2  
m 02  
en 1)  
t [19](#)  
ac re  
hi po  
ev rt  
ed ed  
th an  
ro in  
ug cr  
h ea  
th se  
e in  
m si  
et g  
ho m  
do a  
lo le  
gy ve  
. l  
W fr  
he o  
n m  
se 2.  
t 8  
ag to  
ai 4.  
ns 2  
t af  
si te  
m r  
ila ap  
r pl  
st yi  
ud ng  
ie L  
s, ea  
th n  
e Si  
re x  
su Si  
lts g  
he m



---

a In  
to co  
ac nt  
et ra  
a st,  
m th  
in e  
op cu  
he rr  
n en  
ta t  
bl re  
et se  
pr ar  
od ch  
uc su  
ti rp  
on as  
. se  
T d  
he th  
ir os  
st e  
ud re  
y su  
yi lts  
el wi  
de th  
d a  
ef po  
fic sti  
ie nt  
nc er  
y ve  
ga nt  
in io  
s n  
an si  
d g  
m m  
od a  
er le  
at ve  
e l  
en of  
ha 5.  
nc 4  
e an  
m d  
en a  
ts D  
in P  
qu M  
ali O  
ty. re

du ge  
cti nc  
on y  
ex de  
ce pa  
ed rt  
in m  
g en  
95 t.  
% W  
. hi  
A le  
no i  
th m  
er pr  
ex ov  
a e  
m m  
pl en  
e, ts  
M w  
ist er  
ar e  
ih ac  
i hi  
et ev  
al. ed  
(2 ,  
02 th  
3) e  
,<sup>18</sup> no  
ap ni  
pl nd  
ie us  
d tri  
Si al  
x co  
Si nt  
g ex  
m t  
a m  
te ad  
ch e  
ni th  
qu os  
es e  
in re  
a su  
ho lts  
sp le  
ita ss  
l ap  
e pl  
m ic  
er ab

---

|     |     |
|-----|-----|
| le  | pa  |
| to  | rti |
| m   | cu  |
| an  | la  |
| uf  | rl  |
| ac  | y   |
| tu  | gi  |
| ri  | ve  |
| ng  | n   |
| .   | th  |
| T   | e   |
| he  | co  |
| se  | m   |
| co  | pl  |
| m   | ex  |
| pa  | iti |
| ris | es  |
| on  | of  |
| s   | ph  |
| he  | ar  |
| lp  | m   |
| un  | ac  |
| de  | eu  |
| rs  | tic |
| co  | al  |
| re  | pr  |
| th  | od  |
| e   | uc  |
| str | ti  |
| en  | on  |
| gt  | .   |
| h   | B   |
| an  | ey  |
| d   | on  |
| sc  | d   |
| op  | st  |
| e   | ati |
| of  | sti |
| th  | ca  |
| e   | l   |
| ou  | i   |
| tc  | m   |
| o   | pr  |
| m   | ov  |
| es  | e   |
| ob  | m   |
| se  | en  |
| rv  | ts, |
| ed  | th  |
| in  | e   |
| th  | fin |
| is  | di  |
| w   | ng  |
| or  | s   |
| k,  | re  |

---

|     |     |
|-----|-----|
| fle | d   |
| ct  | ou  |
| m   | tp  |
| ea  | ut  |
| ni  | be  |
| ng  | ca  |
| fu  | m   |
| l   | e   |
| ch  | m   |
| an  | or  |
| ge  | e   |
| s   | pr  |
| in  | ed  |
| ho  | ict |
| w   | ab  |
| th  | le. |
| e   | T   |
| pr  | he  |
| oc  | ou  |
| es  | tc  |
| s   | o   |
| pe  | m   |
| rf  | es  |
| or  | al  |
| m   | so  |
| s   | ca  |
| da  | rr  |
| y-  | y   |
| to  | fin |
| -   | an  |
| da  | ci  |
| y.  | al  |
| D   | i   |
| ef  | m   |
| ec  | pl  |
| t   | ic  |
| ra  | ati |
| te  | on  |
| s   | s,  |
| de  | in  |
| cli | cl  |
| ne  | ud  |
| d,  | in  |
| va  | g   |
| ri  | po  |
| ab  | te  |
| ili | nt  |
| ty  | ial |
| w   | re  |
| as  | du  |
| re  | cti |
| du  | on  |
| ce  | s   |
| d,  | in  |
| an  | w   |

---

|     |     |
|-----|-----|
| as  | re  |
| te  | fe  |
| an  | re  |
| d   | nc  |
| re  | e   |
| w   | po  |
| or  | in  |
| k,  | t   |
| i   | fo  |
| m   | r   |
| pr  | ph  |
| ov  | ar  |
| ed  | m   |
| th  | ac  |
| ro  | eu  |
| ug  | tic |
| hp  | al  |
| ut, | fir |
| an  | m   |
| d   | s   |
| m   | ai  |
| or  | m   |
| e   | in  |
| ef  | g   |
| fic | to  |
| ie  | i   |
| nt  | m   |
| us  | pl  |
| e   | e   |
| of  | m   |
| re  | en  |
| so  | t   |
| ur  | or  |
| ce  | re  |
| s.  | fin |
| Ta  | e   |
| ke  | L   |
| n   | ea  |
| to  | n   |
| ge  | Si  |
| th  | x   |
| er, | Si  |
| th  | g   |
| is  | m   |
| w   | a   |
| or  | str |
| k   | at  |
| se  | eg  |
| rv  | ie  |
| es  | s.  |
| as  | T   |
| a   | he  |
| str | ap  |
| on  | pr  |
| g   | oa  |

ch no  
us t  
ed on  
he ly  
re po  
de ss  
m ib  
on le  
str bu  
at t  
es re  
th pe  
at at  
wi ab  
th le  
str ac  
uc ro  
tu ss  
re ot  
d he  
an r  
al pr  
ys od  
is uc  
an ti  
d on  
fo en  
cu vi  
se ro  
d n  
ac m  
ti en  
on ts.

,  
si  
gn Val  
ifi ida  
ca tio  
nt n  
ga This  
in study  
s prese  
in nted  
qu a  
ali uniqu  
ty e  
an oppor  
d tunity  
ef to  
fic apply  
ie and  
nc valid  
y ate  
ar the  
e

---

Six lendi  
 Sigm ng  
 a validi  
 meth ty  
 odolo and  
 gy in practi  
 a cal  
 real- depth  
 world to the  
 phar proje  
 mace ct.  
 utical By  
 manu follo  
 factur wing  
 ing a  
 envir struct  
 onme ured  
 nt frame  
 throu work  
 gh for  
 direct proce  
 colla ss  
 borati monit  
 on oring,  
 with clearl  
 an y  
 indus establ  
 try- ishin  
 leadi g  
 ng defec  
 comp t  
 any. defini  
 This tions,  
 enga and  
 geme maint  
 nt ainin  
 enabl g  
 ed consi  
 the stenc  
 resea y in  
 rch data  
 team colle  
 to ction  
 work and  
 with analy  
 actua sis  
 l acros  
 produ s all  
 ction DMA  
 data IC  
 and stage  
 opera s, the  
 ting study  
 scena prese  
 rios, rved

---

stron techn  
 g ical  
 intern literat  
 al ure  
 validi stren  
 ty. To gthen  
 broad ed  
 en the  
 exter const  
 nal ruct  
 relev validi  
 ance, ty  
 consi and  
 derati ensur  
 ons ed a  
 were wellr  
 made ound  
 regar ed  
 ding under  
 the stand  
 appli ing of  
 cabili the  
 ty of proce  
 the ss.  
 findin One  
 gs of the  
 beyo resea  
 nd rcher  
 the s was  
 imme activ  
 diate ely  
 case. invol  
 Draw ved  
 ing on-  
 upon site,  
 multi comp  
 ple leting  
 sourc a  
 es, profe  
 inclu ssion  
 ding al  
 histor traini  
 ical ng  
 perfo cours  
 rman e  
 ce withi  
 recor n the  
 ds, same  
 exper comp  
 t any  
 interv respo  
 iews, nsibl  
 and e for  
 relev produ  
 ant cing



---

the theor  
AMO etical  
F625 ly  
tablet justifi  
s. ed  
This but  
proxi also  
mity groun  
allow ded  
ed for in  
regul opera  
ar tional  
intera realit  
ction y.  
with Despi  
key te the  
perso enco  
nnel, uragi  
inclu ng  
ding result  
produ s, the  
ction imple  
mana ment  
gers ation  
and was  
opera not  
tors. witho  
Feed ut  
back chall  
from enges  
these . A  
intera prima  
ction ry  
s was limita  
integr tion  
ated was  
into the  
the short  
proje age  
ct's of  
decisi staff  
ons, famil  
ensur iar  
ing with  
that Six  
the Sigm  
reco a  
mme conce  
nded pts,  
impr whic  
ovem h  
ents requi  
were red  
not additi  
only onal

---

traini hesita  
 ng tion  
 and amon  
 supp g  
 ort to opera  
 ensur tors,  
 e whic  
 align h was  
 ment addre  
 with ssed  
 the throu  
 new gh  
 proce consi  
 dures stent  
 . com  
 More muni  
 over, catio  
 certai n and  
 n gradu  
 techn al  
 ical enga  
 const geme  
 raints nt.  
 — Thes  
 such e  
 as const  
 adapt raints  
 ing ,  
 impr while  
 ovem not  
 ent unus  
 meas ual in  
 ures indus  
 to trial  
 existi envir  
 ng onme  
 mach nts,  
 inery highli  
 — ght  
 requi the  
 red need  
 adjus for  
 tment flexib  
 s le  
 durin plann  
 g the ing  
 Impr and  
 ove stron  
 phase g  
 . colla  
 There borati  
 was on  
 also with  
 some intern  
 initial al

---

teams  
 .  
 Addr  
 essin  
 g  
 them  
 effect  
 ively  
 was  
 essen  
 tial  
 for  
 ensur  
 ing  
 the  
 long-  
 term  
 viabil  
 ity of  
 the  
 impr  
 ovem  
 ents.  
 Finall  
 y, the  
 reliab  
 ility  
 of the  
 meas  
 urem  
 ent  
 syste  
 m  
 was  
 reinf  
 orced  
 throu  
 gh  
 statist  
 ical  
 valid  
 ation  
 using  
 tools  
 like  
 ANO  
 VA  
 and  
 Gage  
 R&R,  
 as  
 detail  
 ed in  
 the  
 Anal

ysis  
 sectio  
 n.  
 Thes  
 e  
 asses  
 smen  
 ts  
 confir  
 med  
 the  
 accur  
 acy  
 of the  
 data  
 and  
 the  
 soun  
 dness  
 of the  
 meas  
 urem  
 ent  
 meth  
 ods.  
 From  
 a  
 perfo  
 rman  
 ce  
 stand  
 point,  
 the  
 impr  
 ovem  
 ents  
 were  
 clear:  
 the  
 sigm  
 a  
 level  
 incre  
 ased  
 from  
 3.046  
 to  
 4.104  
 , and  
 DPM  
 O  
 dropp  
 ed  
 signi  
 ficant

ly Ec  
from on  
1157.  
13 to o  
48.79 mi  
(refer c  
to im  
[Table](#) 4 in pa  
the ct  
Impr an  
ove d  
sectio  
n). co  
Take st  
n ev  
toget  
her, alu  
these ati  
outco  
mes on  
illustr of  
ate D  
both M  
the Al  
meth C  
odolo  
gical ap  
rigor pli  
and ca  
the tio  
practi  
cal n  
effect  
iveness of Appl  
the ying  
Lean the  
Six DMA  
Sigm IC  
a meth  
appro od  
ach initial  
when ly  
appli invol  
ed ved  
under expe  
real nditu  
manu res  
factur relate  
ing d to  
condi traini  
tions. ng  
perso  
nnel,

---

intro overa  
 ducin ll  
 g tablet  
 auto qualit  
 matio y  
 n lower  
 techn ed  
 ologi poten  
 es, tial  
 and expe  
 refini nses  
 ng linke  
 opera d to  
 tional regul  
 practi atory  
 ces. non-  
 Despi comp  
 te lianc  
 these e and  
 upfro mitig  
 nt ated  
 costs, the  
 the risk  
 comp of  
 any costl  
 quick y  
 ly produ  
 recou ct  
 ped recall  
 its s.  
 inves This  
 tment clearl  
 throu y  
 gh under  
 notab score  
 le s the  
 reduc finan  
 tions cial  
 in bene  
 produ fit of  
 ct DMA  
 defec IC  
 ts and adopt  
 dimin ion,  
 ished highli  
 resou ghtin  
 rce g its  
 wasta practi  
 ge. cal  
 More value  
 over, and  
 the econ  
 enha omic  
 ncem viabil  
 ent in ity

withi  
n  
phar  
mace  
utical  
manu  
factur  
ing  
conte  
xts.  
  
Co  
ncl  
usi  
on  
an  
d  
rec  
o  
m  
m  
en  
da  
tio  
ns  
This  
resea  
rch  
delin  
eates  
the  
signi  
ficant  
impa  
ct of  
adopt  
ing  
the  
DMA  
IC  
meth  
odolo  
gy  
assoc  
iated  
with  
Six  
Sigm  
a in  
refini  
ng  
the  
tablet  
comp  
ressio  
n  
proce  
dure  
for  
AMO  
F625  
tablet  
s  
withi  
n the  
phar  
mace  
utical  
indus  
try.  
Thro  
ugh  
comp  
rehen  
sive  
exam  
inatio  
n and  
syste  
matic  
interv  
entio  
ns,  
the  
appli  
catio  
n of  
Six  
Sigm  
a  
yield  
ed  
subst  
antial  
reduc  
tions  
in  
non-  
confo  
rmiti  
es,  
there  
by  
under

---

scoring its effectiveness in fostering operational excellence and improving product quality. Note that LSS constitutes an influential methodological area such as the reliability of measurement systems, identification of root causes, automation of processes, and continuous improvements monitoring through statistical process control (SPC). These outcomes support the claim that LSS constitutes an influential methodological area for enhancing efficiency and quality. However, the inquiry also revealed certain constraints, including challenges related to

---

scala facilities.  
 bility Despi  
 in te  
 larger these  
 manu these  
 factur chall  
 ing enges  
 envir , the  
 onme meth  
 nts odolo  
 and gy  
 poten can  
 tial be  
 resou adapt  
 rce ed  
 limita throu  
 tions, gh a  
 whic phase  
 h d  
 enco imple  
 mpas ment  
 s ation  
 traini strate  
 ng, gy  
 techn and  
 ology by  
 imple optim  
 ment izing  
 ation, the  
 and use  
 huma of  
 n curre  
 capit nt  
 al. resou  
 Such rces.  
 factor Addit  
 s ionall  
 could y, the  
 hinde devel  
 r the opme  
 wides nt of  
 pread intern  
 appli al  
 catio capab  
 n of ilities  
 Six throu  
 Sigm gh  
 a traini  
 meth ng  
 odolo and  
 gies gradu  
 acros al  
 s auto  
 diver matio  
 se n



---

may enhance  
 ease of use  
 the operational  
 transition effi-  
 ciency  
 to be  
 more reduced  
 complexity  
 systems defects and  
 and improving  
 ensure quality  
 long-term  
 viability.  
 The Further  
 concerns  
 points, re,  
 presented in-  
 dustries such  
 as this  
 study medi-  
 cal  
 device  
 applica-  
 ble manu-  
 facturing  
 other and  
 pharmaceutical  
 technology  
 processes, stand-  
 ing to  
 gain subst-  
 antial  
 , ly  
 coating, from  
 and LSS  
 packaging, meth-  
 odologies,  
 where there  
 e by  
 similar enha-  
 ncing  
 meth both  
 odolo opera-  
 tional  
 gies  
 can effect

iveness and productivity. More over, to amplify the advantage of LSS, the incorporation of machine learning for real-time process optimization and predictive maintenance is advocated. Machine learning algorithms possess the capacity to aid manufacturers in the continuous

improvement of production data, prediction of potential failures, and dynamic optimization of processes, ultimately leading to further reduction in defects and improvements in efficiency.<sup>2,3,24</sup>

While this study focuses on the specific

---

all s  
y po  
on te  
th nti  
e al  
ta for  
bl br  
et oa  
co de  
m r  
pr ap  
es pli  
sio cat  
n io  
pr n.  
oc Ot  
es he  
s, r  
th sta  
e ge  
ap s  
pr in  
oa ph  
ch ar  
ta m  
ke ac  
n eu  
— tic  
bo al  
th m  
in an  
ter uf  
ms act  
of uri  
m ng  
et ,  
ho su  
do ch  
lo as  
gy gr  
an an  
d ul  
pr ati  
act on  
ica ,  
l co  
im ati  
pr ng  
ov ,  
e an  
m d  
en pa  
ts ck  
— ag  
ha in

---

g, mi  
als lar  
o str  
fa uc  
ce tur  
si ed  
mi im  
lar pr  
ch ov  
all e  
en m  
ge en  
s t  
rel str  
ate ate  
d gy  
to .  
va Fu  
ria rth  
bil er  
ity m  
, or  
op e,  
er th  
at e  
or pri  
in nc  
flu ipl  
en es  
ce, us  
an ed  
d he  
pr re  
oc ar  
es e  
s no  
co t  
ntr li  
ol. mi  
Th te  
es d  
e to  
ar th  
ea e  
s ph  
co ar  
ul m  
d ac  
be eu  
ne tic  
fit al  
fro in  
m du  
a str  
si y.

---

Se rta  
 ct nt,  
 or m  
 s ay  
 lik als  
 e o  
 m fin  
 ed d  
 ica va  
 l lu  
 de e  
 vi in  
 ce ad  
 s, ap  
 fo tin  
 od g  
 m thi  
 an s  
 uf ap  
 act pr  
 uri oa  
 ng ch  
 , .  
 an Ex  
 d pl  
 ch ori  
 e ng  
 mi th  
 cal es  
 pr e  
 oc po  
 es ssi  
 sin bil  
 g, iti  
 w es  
 he in  
 re fut  
 pr ur  
 eci e  
 sio w  
 n or  
 an k  
 d co  
 co ul  
 nsi d  
 ste he  
 nc lp  
 y cla  
 ar rif  
 e y  
 eq ho  
 ua w  
 lly th  
 im e  
 po m

---

|     |     |
|-----|-----|
| et  | Ac  |
| ho  | kn  |
| ds  | o   |
| ou  | wl  |
| tli | ed  |
| ne  | g   |
| d   | m   |
| in  | en  |
| thi | ts  |
| s   | Th  |
| ca  | is  |
| se  | pa  |
| ca  | pe  |
| n   | r   |
| be  | wa  |
| tai | s   |
| lor | pr  |
| ed  | ep  |
| to  | are |
| fit | d   |
| ot  | wh  |
| he  | ile |
| r   | on  |
| set | sa  |
| tin | bb  |
| gs, | ati |
| co  | cal |
| nsi | lea |
| de  | ve, |
| rin | gr  |
| g   | ant |
| th  | ed  |
| e   | to  |
| sp  | Pr  |
| eci | of. |
| fic | Dr  |
| ne  | .   |
| ed  | M   |
| s   | oh  |
| an  | am  |
| d   | ma  |
| li  | d   |
| mi  | D.  |
| tat | A   |
| io  | L-  |
| ns  | Ta  |
| of  | hat |
| ea  | ,   |
| ch  | fro |
| en  | m   |
| vir | the |
| on  | Un  |
| m   | ive |
| en  | rsi |
| t.  |     |

---

|                                                                                     |                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| ty                                                                                  | 13                                                                                  |
| of                                                                                  | 06                                                                                  |
| Jo                                                                                  | M                                                                                   |
| rd                                                                                  | ais                                                                                 |
| an,                                                                                 | a                                                                                   |
| du                                                                                  | A                                                                                   |
| rin                                                                                 | Ns                                                                                  |
| g                                                                                   | ou                                                                                  |
| th                                                                                  | r  |
| e                                                                                   | htt                                                                                 |
| ac                                                                                  | ps:                                                                                 |
| ad                                                                                  | //o                                                                                 |
| e                                                                                   | rci                                                                                 |
| mi                                                                                  | d.o                                                                                 |
| c                                                                                   | rg/                                                                                 |
| ye                                                                                  | 00                                                                                  |
| ar                                                                                  | -                                                                                   |
| 20                                                                                  | 00                                                                                  |
| 23                                                                                  | 02                                                                                  |
| /2                                                                                  | -                                                                                   |
| 02                                                                                  | 52                                                                                  |
| 4.                                                                                  | 46                                                                                  |
|                                                                                     | -                                                                                   |
| O                                                                                   | 41                                                                                  |
| RC                                                                                  | 44                                                                                  |
| ID                                                                                  |                                                                                     |
| iD                                                                                  | Au                                                                                  |
| s                                                                                   | th                                                                                  |
| M                                                                                   | or                                                                                  |
| oh                                                                                  | co                                                                                  |
| am                                                                                  | nt                                                                                  |
| ma                                                                                  | rib                                                                                 |
| d                                                                                   | uti                                                                                 |
| D                                                                                   | on                                                                                  |
| A                                                                                   | s                                                                                   |
| L-                                                                                  | C                                                                                   |
| Ta                                                                                  | O                                                                                   |
| hat                                                                                 | N                                                                                   |
|  | C                                                                                   |
| htt                                                                                 | EP                                                                                  |
| ps:                                                                                 | TI                                                                                  |
| //o                                                                                 | O                                                                                   |
| rci                                                                                 | N:                                                                                  |
| d.                                                                                  | M                                                                                   |
| or                                                                                  | oh                                                                                  |
| g/                                                                                  | am                                                                                  |
| 00                                                                                  | ma                                                                                  |
| 00                                                                                  | d                                                                                   |
| -                                                                                   | D.                                                                                  |
| 00                                                                                  | A                                                                                   |
| 01                                                                                  |                                                                                     |
| -                                                                                   | L-                                                                                  |
| 64                                                                                  | Ta                                                                                  |
| 03                                                                                  | hat                                                                                 |
| -                                                                                   | co                                                                                  |

---

|     |     |
|-----|-----|
| nc  | d   |
| ei  | en  |
| ve  | sur |
| d   | ed  |
| th  | dat |
| e   | a   |
| res | int |
| ea  | eg  |
| rc  | rit |
| h   | y.  |
| id  | A   |
| ea  | N   |
| an  | A   |
| d   | L   |
| de  | Y   |
| fin | SI  |
| ed  | S   |
| th  | O   |
| e   | F   |
| ov  | D   |
| er  | AT  |
| all | A:  |
| ob  | M   |
| jec | ais |
| tiv | a   |
| es  | A.  |
| of  | Ns  |
| th  | ou  |
| e   | r   |
| stu | an  |
| dy. | aly |
| D   | ze  |
| AT  | d   |
| A   | the |
| C   | dat |
| U   | a,  |
| R   | wh  |
| A   | ile |
| TI  | M   |
| O   | oh  |
| N:  | am  |
| Co  | ma  |
| lle | d   |
| cte | D.  |
| d   | A   |
| by  | L-  |
| M   | Ta  |
| ais | hat |
| a   | pr  |
| A.  | ov  |
| Ns  | ide |
| ou  | d   |
| r   | ov  |
| an  | ers |



---

ig Fu  
ht nd  
an in  
d g  
cri Th  
tic e  
al aut  
ins ho  
ig rs  
hts rec  
du eiv  
rin ed  
g no  
da fin  
ta an  
an cia  
al l  
ysi su  
s. pp  
P ort  
R for  
EP the  
A res  
R ear  
A ch,  
TI aut  
O ho  
N rsh  
O ip,  
F an  
T d/  
H or  
E pu  
M bli  
A cat  
N io  
U n  
S of  
C thi  
RI s  
PT art  
: icl  
M e.  
oh  
a  
m De  
m cl  
ad ar  
D. ati  
A on  
L- of  
Ta co  
ha nfl  
t. ic

tin of  
g thi  
int s  
er art  
es icl  
ts e.  
Th Re  
e fer  
au en  
th ce  
or s  
s 1. P  
de y  
cla z  
re d  
d e  
no k  
po T  
te a  
nti n  
al d  
co K  
nfl e  
ict ll  
s e  
of r  
int P.  
er T  
est h  
wi e  
th si  
res x  
pe -  
ct si  
to g  
th m  
e a  
res h  
ea a  
rc n  
h, d  
au b  
th o  
or o  
shi k  
p, .  
an 5  
d/ t  
or h  
pu e  
bli d  
cat .  
io N  
n

e e  
w l  
e  
Y a  
o n  
r si  
k x  
: si  
M g  
c m  
G a  
r p  
a o  
w c  
- k  
H e  
il t  
l, t  
2 o  
0 o  
0 l  
9 b  
o  
2. G o  
e k  
o .  
r N  
g e  
e w  
M  
L Y  
, o  
R r  
o k  
w :  
l M  
a c  
n G  
d r  
s a  
D w  
, -  
P H  
ri il  
c l,  
e 2  
M 0  
, 0  
e 5  
t .  
a 3. M  
l. a  
T r  
h z

a .  
g I  
a n  
o t  
D J  
a P  
n r  
d o  
C j  
a e  
r c  
v t  
a M  
l a  
h n  
o a  
M g  
. e  
C m  
ri e  
ti n  
c t  
a 2  
l 0  
s 1  
u 6  
c ;  
c 3  
e 4  
s .  
s 4. Y  
f a  
a d  
c a  
t v  
o V  
r .  
s F  
f r  
o a  
r m  
si e  
x w  
si o  
g r  
m k  
a o  
p f  
r i  
o n  
j t  
e e  
c g  
ts r

---

|    |    |
|----|----|
| a  | e  |
| t  | m  |
| e  | e  |
| d  | n  |
| g  | t  |
| r  | a  |
| e  | ti |
| e  | o  |
| n  | n  |
| l  | i  |
| e  | n  |
| a  | m  |
| n  | a  |
| si | n  |
| x  | u  |
| -  | f  |
| si | a  |
| g  | c  |
| m  | t  |
| a  | u  |
| a  | ri |
| n  | n  |
| d  | g  |
| i  | i  |
| d  | n  |
| e  | d  |
| n  | u  |
| ti | st |
| fi | r  |
| e  | y  |
| d  | :  |
| b  | a  |
| a  | c  |
| rr | ri |
| i  | ti |
| e  | c  |
| r  | a  |
| s  | l  |
| f  | li |
| o  | t  |
| r  | e  |
| g  | r  |
| r  | a  |
| e  | t  |
| e  | u  |
| n  | r  |
| l  | e  |
| e  | r  |
| a  | e  |
| n  | v  |
| i  | i  |
| m  | e  |
| p  | w  |
| l  | J  |

E p  
n e  
g r  
A a  
p ti  
p o  
l n  
S s  
c m  
i a  
2 n  
0 a  
1 g  
9 e  
; m  
5 e  
4 n  
: t:  
3 p  
5 r  
5 o  
. c  
5. K e  
r s  
a s  
j e  
e s  
w a  
s n  
k d  
i v  
L a  
J, l  
R u  
it e  
z c  
m h  
a a  
n i  
L n  
P s.  
a 1  
n 0  
d t  
M h  
a e  
l d  
h .  
o P  
tr r  
a e  
M n  
. ti  
O c

---

e d  
H t  
a h  
ll e  
, w  
2 o  
0 rl  
1 d  
6 .  
. I  
6. W n  
o :  
m T  
a h  
c e  
k st  
J o  
P, r  
J y  
o o  
n f  
e l  
s e  
D a  
T n  
a p  
n r  
d o  
R d  
o u  
o c  
s ti  
D o  
. n  
T T  
h o  
e y  
m o  
a t  
c a'  
h s  
i s  
n e  
e c  
t r  
h e  
a t  
t w  
c e  
h a  
a p  
n o  
g n  
e i

n ,  
t N  
h Y  
e ;  
g S  
l i  
o m  
b o  
a n  
l a  
c n  
a d  
r S  
w c  
a h  
r u  
s st  
t e  
h r,  
a 1  
t 9  
is 9  
r 1  
e .  
v 7. A  
o l-  
l T  
u a  
ti h  
o a  
n t  
i M  
z D  
i .  
n E  
g ff  
w e  
o c  
rl ti  
d v  
i e  
n d  
d e  
u si  
st g  
r n  
y. a  
N n  
e d  
w a  
y n  
o a  
r l  
k y



---

|    |      |
|----|------|
| si | 2    |
| s  | 0    |
| o  | 1    |
| f  | 0    |
| p  | ;    |
| a  | 1    |
| tt | 0    |
| e  | (    |
| r  | 1    |
| n  | 1    |
| m  | ):   |
| a  | 8    |
| k  | 7    |
| i  | 8    |
| n  | -    |
| g  | 8    |
| p  | 8    |
| r  | 6.   |
| o  | 8. A |
| c  | l-   |
| e  | T    |
| s  | a    |
| s  | h    |
| u  | a    |
| si | t    |
| n  | M    |
| g  | D    |
| v  | ,    |
| a  | A    |
| l  | l-   |
| u  | R    |
| e  | e    |
| st | f    |
| r  | a    |
| e  | i    |
| a  | e    |
| m  | A    |
|    | a    |
| m  | n    |
| a  | d    |
| p  | A    |
| p  | l-   |
| i  | D    |
| n  | w    |
| g  | a    |
| .  | ir   |
| J  | i    |
| A  | A    |
| P  | F    |
| P  | .    |
| l  | P    |
| S  | e    |
| c  | rf   |
| i  | o    |
|    | r    |
|    | m    |

a p  
n li  
c n  
e g  
e i  
v n  
a s  
l p  
u e  
a c  
ti ti  
o o  
n n  
a ,  
n Ij  
d is  
a e  
n 2  
a 0  
l 1  
y 2  
si ;  
s 1  
o 1  
f (  
a 3  
J ):  
I 2  
T 2  
- 5  
K -  
a 2  
n 4  
b 9  
a .  
n 9. A  
p l-  
r T  
o a  
d h  
u a  
c t  
ti M  
o D  
n .  
s O  
y p  
st ti  
e m  
m i  
  
w z  
it i  
h n  
s g  
a o  
m f

---

w E  
o n  
r g  
k S  
i c  
n i  
p 2  
r 0  
o 0  
g 5  
r ;  
e 3  
s 2  
s (  
( 2  
W ):  
I 1  
P 2  
) 3  
i -  
n 1  
K 3  
a 2  
n ,  
b T  
a h  
n e  
c u  
o n  
n i  
tr v  
o e  
ll r  
e si  
d t  
p y  
r o  
o f  
d J  
u o  
c r  
ti d  
o a  
n n  
li .  
n [  
e O  
s. n  
D li  
ir n  
a e  
s ].  
a A  
t, v

a 9  
il f  
a 1  
b 4  
l 0  
e e  
: 2  
h 4  
tt f  
p 9  
s: 7  
// f  
p 9  
d 8  
f f  
s. b  
s a  
e 3  
m f  
a e  
n 4  
ti .  
c p  
s d  
c f  
h 10. A  
o l-  
l T  
a a  
r. h  
o a  
r t  
g M  
/ D  
1 .  
d A  
e n  
9 a  
/ l  
2 y  
f si  
7 s  
6 o  
5 f  
0 t  
0 h  
e e  
c d  
d is  
f r  
2 u  
f p  
8 ti  
b o

---

|    |    |
|----|----|
| n  | u  |
| s  | s  |
| o  | i  |
| f  | m  |
| t  | p  |
| h  | r  |
| e  | o  |
| d  | v  |
| i  | e  |
| g  | m  |
| it | e  |
| a  | n  |
| lr | t  |
| e  | [  |
| v  | 2  |
| o  | 0  |
| l  | ]  |
| u  | m  |
| ti | e  |
| o  | t  |
| n  | h  |
| a  | o  |
| n  | d  |
| d  | o  |
| it | l  |
| s  | o  |
| i  | g  |
| m  | i  |
| p  | e  |
| a  | s  |
| c  | a  |
| ts | n  |
| o  | d  |
| n  | l  |
| t  | e  |
| h  | a  |
| e  | n  |
| e  | si |
| v  | x  |
| o  | -  |
| l  | si |
| u  | g  |
| ti | m  |
| o  | a. |
| n  | Ij |
| o  | ir |
| f  | s  |
| c  | s  |
| o  | 2  |
| n  | 0  |
| ti | 2  |
| n  | 4  |
| u  | ;  |
| o  | 7  |

( x  
4 si  
): g  
1 m  
5 a  
6 m  
3 e  
- t  
1 h  
5 o  
7 d  
9 o  
. l  
11. P o  
a g  
n y  
d :  
y a  
a m  
B o  
D d  
, e  
D r  
i n  
w a  
a p  
k p  
a r  
r o  
S a  
A c  
a h  
n i  
d n  
S p  
a h  
l a  
u r  
j m  
a a  
A c  
K e  
. u  
A ti  
r c  
e a  
v l  
i m  
e a  
w n  
o f  
n a  
si c

---

t 12. A  
u l  
ri h  
n u  
g r  
i a  
n is  
d h  
u I,  
st R  
r o  
y. b  
W l  
o e  
rl d  
d o  
J. C  
P a  
h n  
a d  
r K  
m o  
. b  
P i  
h A  
a .  
r A  
m c  
. o  
S m  
c p  
i a  
2 r  
0 a  
2 ti  
2 v  
; e  
1 e  
1 x  
( p  
1 l  
0 o  
): r  
1 a  
8 ti  
4 o  
6 n  
- o  
1 f  
8 l  
6 e  
1 a  
. n  
m

a C  
n l  
u e  
f a  
a n  
c P  
t r  
u o  
ri d  
n 2  
g 0  
a 1  
n 7  
d ;  
si 1  
x 6  
si 4  
g :  
m 3  
a 2  
i 5  
n -  
t 3  
e 3  
r 7  
m .  
s 13. A  
o l-  
f T  
t a  
h h  
e a  
ir t  
c M  
ri D  
ti a  
c n  
a d  
l M  
s o  
u h  
c a  
c m  
e m  
s e  
s d  
f M  
a J.  
c L  
t e  
o a  
r n  
s. -  
J b



---

|    |    |
|----|----|
| a  | p  |
| s  | r  |
| e  | o  |
| d  | a  |
| d  | c  |
| e  | h  |
| si | f  |
| g  | o  |
| n  | r  |
| a  | J  |
| n  | o  |
| d  | r  |
| a  | d  |
| n  | a  |
| a  | n  |
| l  | i  |
| y  | a  |
| si | n  |
| s  | l  |
| o  | a  |
| f  | n  |
| a  | d  |
| c  | tr |
| o  | a  |
| n  | n  |
| ti | s  |
| n  | p  |
| u  | o  |
| o  | rt |
| u  | i  |
| s  | n  |
| i  | d  |
| m  | u  |
| p  | st |
| r  | r  |
| o  | y. |
| v  | Ij |
| e  | m  |
| m  | st |
| e  | 2  |
| n  | 0  |
| t  | 2  |
| e  | 3  |
| n  | ;  |
| g  | 1  |
| i  | 0  |
| n  | (  |
| e  | 2  |
| e  | ): |
| ri | 3  |
| n  | 5  |
| g  | 9  |
| a  | 9  |
| p  | -  |

3 p  
6 r  
0 o  
0 a  
. c  
14. A h  
l- t  
R o  
e m  
f e  
a a  
i s  
e u  
A r  
, e  
A p  
l- r  
T o  
a d  
h u  
a c  
t ti  
M v  
D it  
a y  
n c  
d h  
N a  
a n  
j g  
d e  
a o  
w f  
i a  
R J  
. o  
U r  
si d  
n a  
g n  
m i  
a a  
l n  
m c  
q o  
u m  
is p  
ti a  
n n  
d y  
e f  
x o  
a r  
p p

---

l d  
 a A  
 st b  
 i u  
 c N  
 i a  
 n h  
 d l  
 u e  
 st h  
 ri Y  
 e L  
 s. .  
 A O  
 m r  
 J d  
 O i  
 p n  
 e a  
 r ll  
 R o  
 e g  
 s; is  
 5 ti  
 : c  
 3 r  
 8 e  
 4 g  
 - r  
 4 e  
 0 s  
 0 si  
 . o  
 15. A n  
 l- m  
 T o  
 a d  
 h e  
 a l  
 t o  
 M f  
 D f  
 , a  
 K il  
 a u  
 r r  
 e e  
 e m  
 m o  
 d  
 M e  
 A a  
 a n  
 n d

e a  
ff l  
e 2  
c 0  
ts 1  
a 3  
n ;  
a 2  
l 7  
y :  
si 3  
s 2  
( 2  
F -  
M 3  
E 3  
A 2  
) .  
i 16. A  
n L  
p -  
h T  
a a  
r h  
m a  
a t  
c M  
e D  
u .  
ti D  
c i  
a g  
l it  
t a  
a l  
b T  
l e  
e c  
ti h  
n n  
g o  
t l  
o o  
o g  
ls i  
. e  
E s  
n a  
g n  
F d  
a t  
il h  
A e  
n E

---

v n  
o E  
l n  
u g  
ti i  
o n  
n e  
o e  
fl ri  
n n  
d g  
u a  
st n  
r d  
y S  
4 c  
. i  
0 e  
a n  
n c  
d e,  
A B  
g e  
il t  
it w  
y. e  
T e  
h n  
e 9  
6<sup>t</sup> -  
h 1  
I 3  
n ,  
t N  
e o  
r v  
n e  
a m  
ti b  
o e  
n r  
a 2  
l 0  
C 2  
o 3  
n i  
f n  
e A  
r n  
e t  
n a  
c l  
e y  
o a,

T t  
u a  
r l,  
k W  
e a  
y, t  
i e  
n r  
c a  
o n  
ll d  
a D  
b e  
o s  
r e  
a rt  
ti R  
o e  
n s  
w e  
it a  
h r  
E c  
u h  
r a  
o n  
- d  
A W  
r a  
a si  
b t  
O U  
r n  
g i  
a v  
n e  
i r  
z si  
a t  
ti y  
o /  
n Ir  
f a  
o q  
r ,  
E 2  
n 0  
v 2  
ir 3  
o .  
n 17. A  
m l-  
e T  
n a

---

|    |    |
|----|----|
| h  | e  |
| a  | q  |
| t  | u  |
| M  | i  |
| D  | p  |
| a  | m  |
| n  | e  |
| d  | n  |
| A  | t  |
| l- | m  |
| R  | a  |
| if | i  |
| a' | n  |
| e  | t  |
| F  | e  |
| .  | n  |
| A  | a  |
| p  | n  |
| p  | c  |
| l  | e  |
| y  | m  |
| i  | a  |
| n  | n  |
| g  | a  |
| a  | g  |
| d  | e  |
| a  | m  |
| t  | e  |
| a  | n  |
| a  | t  |
| n  | t  |
| a  | o  |
| l  | i  |
| y  | m  |
| ti | p  |
| c  | r  |
| s  | o  |
| a  | v  |
| p  | e  |
| p  | t  |
| r  | h  |
| o  | e  |
| a  | m  |
| c  | e  |
| h  | a  |
| t  | n  |
| o  | ti |
| m  | m  |
| e  | e  |
| d  | b  |
| i  | e  |
| c  | t  |
| a  | w  |
| l  | e  |

e 18. M  
n is  
f t  
a a  
il ri  
u h  
r i  
e M  
a Z  
n ,  
d A  
a l-  
v T  
a a  
il h  
a a  
b t  
il M  
it D  
y. a  
J n  
o d  
r A  
d l-  
a N  
n i  
J m  
M e  
e r  
c S  
h H  
I .  
n I  
d m  
E p  
n r  
g o  
2 v  
0 i  
2 n  
3 g  
; p  
1 r  
7 o  
( c  
1 e  
): s  
3 s  
3 e  
- f  
4 fi  
0 c  
. i  
e



---

n si  
c x  
y -  
a si  
t g  
p m  
e a  
d si  
i m  
a u  
tr l  
i a  
c ti  
h o  
o n  
s m  
p e  
it t  
a h  
l o  
e d  
m o  
e l  
r o  
g g  
e y.  
n P  
c r  
y o  
d c  
e e  
p s  
a s  
rt e  
m s  
e 2  
n 0  
t 2  
u 3  
si ;  
n 1  
g 1  
a (  
n 2  
i ):  
n 3  
t 9  
e 9  
g .  
r 19. B  
a y  
t r  
e n  
d e

B h  
, a  
M r  
c m  
D a  
e c  
r e  
m u  
o ti  
tt c  
O a  
a l  
n m  
d a  
N n  
o u  
o f  
n a  
a c  
n t  
J. u  
A ri  
p n  
p g  
l f  
y a  
i c  
n il  
g it  
l y  
e :  
a a  
n c  
si a  
x s  
si e  
g st  
m u  
a d  
m y.  
e P  
t r  
h o  
o c  
d e  
o s  
l s  
o e  
g s  
y 2  
t 0  
o 2  
a 1  
p ;

---

9 t  
 ( h  
 3 e  
 ): p  
 5 h  
 5 a  
 0 r  
 . m  
 20. B a  
 e c  
 ll e  
 E u  
 , ti  
 V c  
 o a  
 s l  
 s i  
 e n  
 n d  
 B u  
 , st  
 T r  
 o y.  
 b I  
 i n  
 n :  
 E P  
 , r  
 e o  
 t c  
 a e  
 l. e  
 T d  
 h i  
 e n  
 e g  
 f 2  
 fi 0  
 c 2  
 i 3  
 e I  
 n n  
 c t  
 y e  
 o r  
 f n  
 si a  
 x ti  
 si o  
 g n  
 m a  
 a l  
 i C  
 n o

n C  
f S  
e C  
r I)  
e .  
n L  
c a  
e s  
C V  
o e  
m g  
p a  
u s,  
t N  
a V  
ti ,  
o U  
n S  
a A  
l ,  
S 1  
c 3  
i -  
e 1  
n 5  
c D  
e e  
a c  
n e  
d m  
C b  
o e  
m r  
p 2  
u 0  
t 2  
a 3  
ti ,  
o p  
n p  
a .  
l 4  
I 3  
n 8  
t -  
e 4  
ll 4  
i 2  
g .  
e 21. P  
n ri  
c n  
e c  
( i

---

|    |    |
|----|----|
| p  | n  |
| l  | e. |
| e  | c  |
| o  | o  |
| f  | m  |
| t  | /  |
| a  | 2  |
| b  | 0  |
| l  | 1  |
| e  | 6  |
| t  | /  |
| c  | 0  |
| o  | 2  |
| m  | /  |
| p  | p  |
| r  | ri |
| e  | n  |
| s  | c  |
| si | i  |
| o  | p  |
| n  | l  |
| m  | e  |
| a  | -  |
| c  | o  |
| h  | f- |
| i  | t  |
| n  | a  |
| e, | b  |
| A  | l  |
| v  | e  |
| a  | t  |
| il | c  |
| a  | o  |
| b  | m  |
| l  | p  |
| e  | r  |
| ∴  | e  |
| h  | s  |
| tt | si |
| p  | o  |
| s: | n  |
| // | -  |
| w  | m  |
| w  | a  |
| w  | c  |
| .  | h  |
| p  | i  |
| h  | n  |
| a  | e. |
| r  | h  |
| m  | t  |
| a  | m  |
| g  | l, |
| u  | A  |
| i  | c  |
| d  | c  |
| e  | e  |
| li | s  |

s e  
e c  
d t  
M o  
a f  
y r  
2 e  
1 li  
, a  
2 b  
0 il  
2 it  
4 y  
. a  
22. W n  
i d  
d d  
a e  
g m  
d p  
o a  
S t  
, h  
H y  
a o  
n n  
d c  
a u  
y st  
a o  
n m  
i e  
Y r  
I, s  
P a  
r ti  
a s  
st f  
y a  
o c  
w ti  
a o  
ti n  
A :  
H a  
, s  
e u  
t r  
a v  
l. e  
T y  
h o  
e f  
e P  
ff T

---

t 8  
 e 1  
 l -  
 k 1  
 o 9  
 m 4  
 I .  
 n 23. T  
 d a  
 o o  
 n C  
 e J,  
 si C  
 a' h  
 s e  
 I n  
 n S  
 d C  
 i a  
 H n  
 o d  
 m C  
 e h  
 c e  
 u n  
 st K  
 o S  
 m .  
 e T  
 r h  
 s. e  
 H st  
 u u  
 m d  
 y  
 S o  
 y f  
 st m  
 M u  
 a lt  
 n ii  
 a n  
 g t  
 2 e  
 0 ll  
 2 i  
 4 g  
 ; e  
 4 n  
 3 c  
 ( e  
 2 e  
 ): v  
 1 a

l f  
u a  
a c  
ti t  
o u  
n ri  
o n  
n g  
t .  
h H  
e u  
a m  
p  
p S  
li y  
c st  
a M  
ti a  
o n  
n a  
o g  
f 2  
v 0  
o 0  
c 9  
a ;  
ti 2  
o 8  
n (  
a 1  
l -  
a 2  
p ):  
ti 6  
t 3  
u -  
d 7  
e 5  
- .  
a 24. B  
n u  
e s  
x h  
a D  
m H  
p a  
l n  
e d  
o D  
n o  
m o  
a l  
n e  
u y



---

K n  
J. t  
A m  
l a  
e n  
a a  
r g  
n e  
i m  
n e  
g n  
p t.  
r H  
o u  
c m  
e  
s S  
s y  
f st  
o M  
r a  
tr n  
a a  
n g  
s l  
f 9  
o 9  
r 2  
m ;  
a 1  
ti 1  
o (  
n 4  
t );  
o 1  
c 8  
o 1  
n -  
ti 1  
n 9  
u 2  
o .  
u  
s  
i  
m  
p  
r  
o  
v  
e  
m  
e